



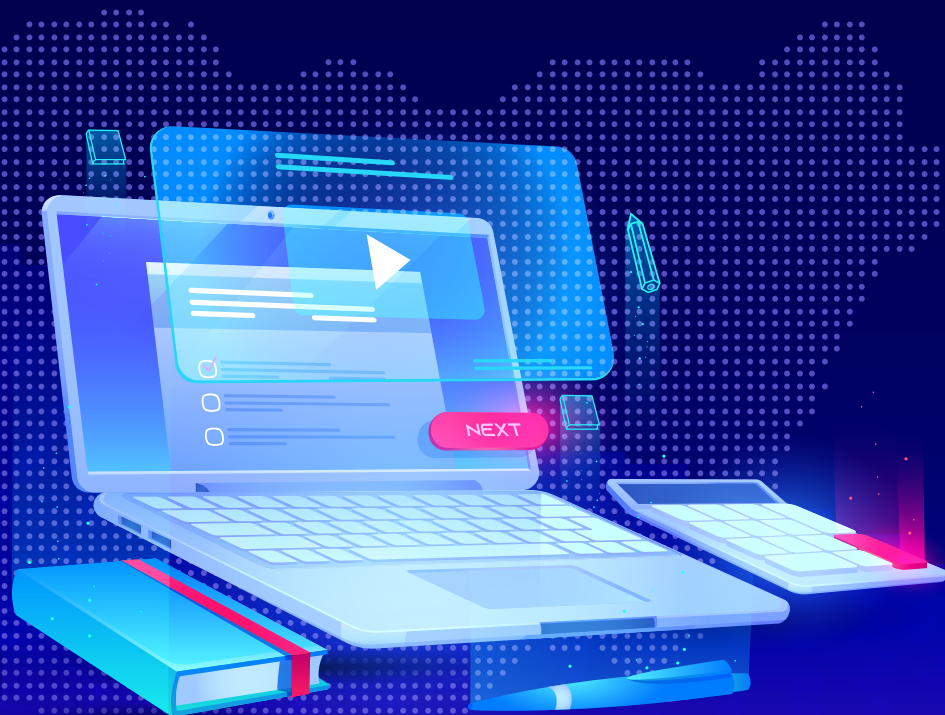
Patrick J McGovern
FOUNDATION



BRAIN BUILDERS
YOUTH DEVELOPMENT
INITIATIVE

Towards an AI-Enabled
Education System in Nigeria:

A Framework for Inclusive and Ethical Integration



Towards an AI-Enabled
Education System in Nigeria:

A Framework for Inclusive and Ethical Integration

WITH SUPPORT FROM:



Patrick J McGovern
F O U N D A T I O N

October 2025

Acknowledgement

This report was produced by Brain Builders Youth Development Initiative with support from Patrick J McGovern Foundation.

Disclaimer

This research is published by Brain Builders Youth Development Initiative (BBYDI) for informational purposes only. Although every effort has been made to ensure accuracy, BBYDI does not guarantee completeness or applicability across all contexts. The views expressed are those of the authors and do not necessarily represent BBYDI or its partners. This publication is not a substitute for professional advice and BBYDI disclaims liability for actions taken based on its content. Reproduction or distribution is permitted with attribution to BBYDI.

©Brain Builders Youth Development Initiative 2025.

BBYDI Secretariat

Email: brainbuilderedu@gmail.com

Tel: 07068775529

Nigeria Address: No. 12 T.O.S. Benson Crescent, Utako Abuja, 900108, FCT

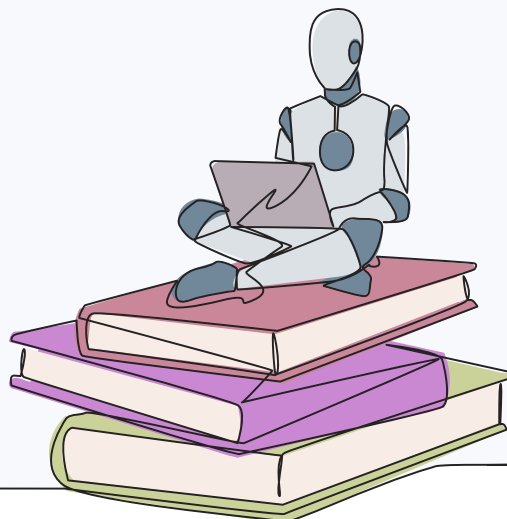
www.thebrainbuilders.org

WITH SUPPORT FROM:



Patrick J McGovern
FOUNDATION

Key Messages



AI can significantly transform education in Nigeria: AI has the potential to improve learning outcomes through personalised learning, automated assessment and data-driven decision-making. It is especially valuable in a context where teacher shortages, quality gaps and inequities persist.

Nigeria has strong policy momentum but early-stage implementation: National strategies now recognise AI as essential for human-capital development and curriculum reform is underway to integrate AI, coding and robotics. However, deployment remains fragmented and infrastructure gaps risk uneven benefits.

Teacher readiness is the key determinant of success: Most teachers are willing but lack training and access to AI tools, which limits classroom integration. Systematic professional development and support structures must accompany technology rollout.

Infrastructure constraints threaten inclusion and scale: Connectivity, power supply, devices and digital access remain weak, especially in rural and low-income communities. Without addressing these gaps, AI adoption may widen educational inequalities.

Curriculum reforms are progressing but require deeper AI literacy and ethics: New curriculum elements now include digital literacy and introductory AI modules. Yet deeper integration, including computational thinking, machine learning foundations, and AI ethics, is not widespread across levels.

Girls and marginalised learners face a higher risk of being left behind: Gender digital divides and online safety concerns limit girls' access to AI-enabled tools and learning opportunities. Inclusion must be intentionally designed and monitored in all AI-in-education initiatives.

Nigeria needs ethical safeguards and governance frameworks: As AI is introduced into classrooms, issues of privacy, data protection and responsible use must be addressed. Governance should ensure AI enhances, rather than replaces, human judgment and learner rights.

Sustainable financing and partnerships are essential for scale: Government investment alone is insufficient to support nationwide AI adoption. Collaboration with private sector, edtech, and global development partners can accelerate innovation and long-term sustainability.

Executive Summary

Nigeria stands at a significant moment where Artificial Intelligence (AI) can accelerate progress toward inclusive, high-quality education for all learners. As global education systems increasingly leverage AI for personalised learning, intelligent tutoring, learner analytics, teacher support, and data-driven administration, the risks of falling behind are high. This report provides a national framework for structured, ethical and equitable integration of AI into Nigeria's education sector, aligned with national development priorities and global standards such as UNESCO's AI Ethics Guidelines and the SDG4 agenda.

The report highlights major opportunities: improving literacy and foundational learning,

expanding reach into underserved communities, strengthening teacher effectiveness, and equipping learners with future-ready digital and AI skills. Emerging Nigerian pilots, such as the Edo State AI tutoring programme, already demonstrate significant learning gains, particularly for girls, and affirm the potential for responsible AI at scale. However, integration remains nascent and uneven, with systemic challenges including limited digital infrastructure, teacher readiness gaps, insufficient curriculum alignment, cost and access barriers for rural and low-income learners, and risks of widening gender and disability divides.

To ensure AI becomes a tool for equity rather than exclusion, the Framework proposes six strategic objectives:



Build AI literacy for students, teachers, and administrators

Promote equitable access and participation across demographics and regions

Strengthen teacher capacity and pedagogical transformation

Embed human-centred, transparent and ethical AI governance

Accelerate local innovation and sustainable adoption

Improve learning outcomes and support data-driven decision-making

The Framework adopts a Theory of Change that positions AI as an enabler of improved learning outcomes only when curriculum updates, teacher training, technology infrastructure, responsible governance, and inclusive financing evolve together. Implementation will require strong multi-stakeholder collaboration across government, education institutions, industry and civil

society, phasing investments over the next 5–10 years.

If executed effectively, this agenda will not only raise learning outcomes but also build Nigeria's workforce competitiveness and ensure young people, especially girls and marginalised learners, can fully participate in an increasingly AI-driven global economy.



Table of Contents

Key Messages	04	5.0 AI Education Across Different Learning Levels	27
Executive Summary	05	Early Childhood & Primary Education	28
Abbreviation	07	Secondary Education	29
1.0 General Introduction	08	Tertiary & Vocational Education	32
Background on AI and its Transformative Potential for Nigeria's Education Sector	9	Non-Formal & Adult Education	35
Purpose, Scope and Vision of the AI in Education	10	Girls Education & Inclusion Across All Levels	37
Strategic Objectives	11		
2.0 Research Methodology	13	6.0 Teacher Capacity Building	39
Methodology	14	7.0 Infrastructure and Equitable Access	43
Theory of Change	14	8.0 AI Content and Resource Development	48
Limitation	14	9.0 Ethical, Legal, and Governance Framework	52
Theoretical Framework: Alignment with National Development Priorities	14	10. Monitoring, Evaluation, & Research	57
3.0 Programme and Policy Frameworks on AI Education in Nigeria	16	11. Financing and Partnerships	62
Current AI, Education Policies and Laws in Nigeria	17	12. Implementation Roadmap and Phasing	66
Current Education Programmes on AI Education in Nigeria	18	Conclusion	68
International Instruments on Inclusive AI in Education	19	References	69
4.0 Nigeria's Education Curriculum and AI Integration	23		

Abbreviation

ICT	Information and Communication Technology
TVET	Technical and Vocational Education and Training
ODFeL	Open, Distance, Flexible & e-Learning
OER	Open Educational Resources
NPE	National Policy on Education
NDEPS	National Digital Economy Policy and Strategy
NITDA	National Information Technology Development Agency
NAIS	National Artificial Intelligence Strategy
UNESCO	United Nations Educational, Scientific and Cultural Organization
OECD	Organisation for Economic Co-operation and Development
SDG 4	Sustainable Development Goal 4
PD	Professional Development (teachers)
KPIs	Key Performance Indicators
DPIA	Data Protection Impact Assessment
NDPA	Nigeria Data Protection Act
NDPC	Nigeria Data Protection Commission
SES	Socio-Economic Status
LAYS	Learning-Adjusted Years of Schooling
LMS	Learning Management System
3MTT	3 Million Technical Talent (Programme)
EdTech	Education Technology
USPF	Universal Service Provision Fund
NMEC	National Mass Education Commission

1.

General Introduction

Background on AI and its Transformative Potential for Nigeria's Education Sector

Artificial Intelligence (AI) refers to systems or machines that mimic human intelligence to perform tasks and can iteratively improve themselves based on the data they collect. Broadly speaking, AI encompasses technologies such as machine learning, natural-language processing, computer vision, and expert systems. In the education sector, AI is increasingly employed for adaptive / personalized learning, intelligent tutoring systems, automated assessment and feedback, learning analytics, and administrative automation.¹ The global education-technology landscape is being reshaped by AI: classrooms are shifting from one-size-fits-all models toward tailored learning experiences, teachers are being supported with AI-driven tools, and educational systems are exploring efficiencies in assessment, monitoring and student support.²

*The importance of foundational AI literacy among young Nigerians is well illustrated in the Nigerian AI literacy storybook, **The Master not the Machine**, where students learn that AI depends on data and still requires human judgment.³*

Artificial Intelligence (AI) is reshaping how education can be delivered, accessed, and managed around the world. In the Nigerian context, this transformation holds particular promise given persistent challenges in access, equity, teacher capacity, infrastructure and learning outcomes.

AI can enable personalised learning paths, adaptive assessment and intelligent tutoring systems which adjust to each learner's needs; free up teacher time from routine tasks; facilitate

resource-efficient delivery (for example via chatbots or offline AI tutors); and support data-driven policy and practice by analysing student performance and curriculum gaps. For Nigeria, these affordances are particularly significant given the large youth population, high unmet educational needs (especially in underserved, rural and internally displaced populations) and the urgent need to accelerate learning outcomes. This is acknowledged by the minister education recently when he said.

"The world is currently experiencing a technological revolution, with Artificial Intelligence at its centre. AI is no longer a vision of the future but a present reality that is reshaping industries, economies, and societies. Integrating AI into teaching provides a unique opportunity to enrich learning, personalise instruction, and build critical thinking skills in our students. Our ultimate aim is to equip them for careers in a technology-driven world, where AI literacy will be as essential as basic reading and writing."⁴

Maruf Olatunji Alausa
Minister of Education, 2025

In a landmark intervention conducted in Edo State, Nigeria during June–July 2024, about 800 first-year senior-secondary students attended twice-weekly after-school English classes in computer labs, where they used a generative AI tool (Microsoft Copilot powered by ChatGPT) under teacher supervision. According to a blog by World Bank, the students who used the AI tutor significantly outperformed their peers in the control group, by roughly 0.30 standard deviations, which the authors equate to about 1.5–2 years of typical learning gain.⁵ Importantly, the gains extended into students' regular year-end curricular exams (beyond the six-week English focus) and included girls, who were initially lagging gaining even more from the intervention.⁶ The pilot

¹ https://www.researchgate.net/publication/390163119_AI_in_education_A_review_of_personalized_learning_and_educational_technology/

² https://www.researchgate.net/publication/390163119_AI_in_education_A_review_of_personalized_learning_and_educational_technology

³ <https://drive.google.com/file/d/1qGjrtkp1Z2n25A2i8XNBCpZUqw9L7xW1/view>

⁴ https://exwhyzee.ng/Bpg_info/28/-Federal-Government-introduces-artificial-intelligence-training-for-6-000-educators

⁵ <https://blogs.worldbank.org/en/developmenttalk/addressing-the-learning-crisis-with-generative-ai--lessons-from->

demonstrates that, in a context of constrained resources, a well-designed generative AI intervention, coupled with strong teacher facilitation and monitoring can deliver meaningful learning improvements in Nigeria's secondary education system.

Research shows that Nigerian students who used AI systems reported positive impacts on their academic performance; one study found that approximately 90% of respondents reported positive benefit. A survey-based study of Nigerian tertiary-level students found that of 509 respondents across the country, 457 students (89.7 %, i.e., approximately 9 in 10) said that use of artificial intelligence tools had a positive impact on their educational performance.⁷ Specifically, respondents noted improvements such as deeper subject knowledge (26.3 % of respondents), enhanced learning and study skills (24.6 %), quicker understanding of technical topics (19.8 %), and stronger language skills (4.1%).⁸ The study also found a small proportion (5.3%) reported negative influences such as decreased reading habits, and 3.7 % indicated they had become lazy due to AI reliance.⁹ These findings suggest strong student-perceived benefits from AI tools in Nigerian higher-education settings, yet also raise caution about potential side-effects and the need for structured, ethical deployment.

Reports on the regional and national AI market indicate that the AI sector in Africa and particularly in Nigeria is expanding swiftly. One summary of Africa's AI market estimates a compound annual growth rate (CAGR) of 27.4 % from 2025 to 2030, increasing from US\$4.51 billion to \$16.53 billion across the continent.¹⁰ For Nigeria specifically, one study by the Centre for the Study of the Economies of Africa (CSEA) projects the domestic AI market will reach US\$434.4 million by 2026.¹¹ Given this growth trajectory, the argument is made that the education sector cannot lag behind: to leverage AI's transformative potential for productivity, innovation and competitiveness, Nigeria must embed AI-capable education

systems, teacher capacity, digital infrastructure and governance around AI in education if it is to capture its full benefit.

1.2

Purpose, Scope and Vision of the AI in Education

Purpose: The purpose of this Framework is to provide a structured national approach for the integration of AI into Nigeria's education system in a way that enhances learning outcomes, strengthens teacher capacity, supports inclusive access, and aligns with national development priorities. The Framework is designed for policy makers, education institutions (primary, secondary, tertiary), teachers, students, EdTech providers and relevant stakeholders.

Scope: This Framework covers:

- All levels of formal education (basic, secondary, tertiary) and non-formal learning contexts.
- Key functions of the education ecosystem: curriculum and pedagogy, teacher professional development, learner assessment, infrastructure and technology, governance and ethics, financing and partnerships.
- Integration of AI tools and systems (e.g., intelligent tutoring, adaptive learning platforms, learning analytics, curriculum design AI, teacher support tools) with an emphasis on equity, inclusion (including gender and learners with special needs), and alignment with national policy.
- Temporal scope: a medium-term horizon (e.g., next 5–10 years) for phased implementation, scale-up and sustainable integration.

⁶ <https://blogs.worldbank.org/en/education/From-chalkboards-to-chatbots-Transforming-learning-in-Nigeria>

⁷ <https://systems.enpress-publisher.com/index.php/jipd/article/viewFile/9949/5250>

⁸ <https://systems.enpress-publisher.com/index.php/jipd/article/viewFile/9949/5250>

⁹ <https://systems.enpress-publisher.com/index.php/jipd/article/viewFile/9949/5250>

¹⁰ <https://www.ecofinagency.com/news-digital/1308-48038-africa-s-ai-market-poised-to-reach-16-5-billion-by-2030-mastercard-reports>

¹¹ <https://www.arise.tv/csea-nigerias-ai-market-to-hit-434-4m-by-2026-but-adoption-remains-uneven/>

Box 1: Vision

"A Nigeria where every learner, regardless of location or background, benefits from AI-enabled, personalised, inclusive and high-quality education; where teachers are empowered by AI tools; and where the education system uses data and intelligence to close learning gaps, build future-ready skills and support national development."

13

Strategic Objectives

The strategic objectives of this framework set the foundational goals that will steer the adoption of artificial intelligence (AI) across Nigeria's education system. These objectives are designed to ensure that AI becomes a force for quality learning, equitable access, teacher professionalism, and ethical innovation. They are aligned with national development priorities and global best practices in responsible AI, while being tailored to Nigeria's unique context of diverse learners, variable infrastructure, and the need to build future-ready human capital.

Objective 1: Build AI Literacy and Learning-Readiness

Equip students, teachers and educational administrators with the knowledge, skills and mind-sets required to engage confidently with AI-driven learning environments. This objective emphasises not only foundational digital and data literacy, but also AI-specific competencies: understanding what AI is and how it works, being able to use AI-based tools for learning, and developing the critical thinking and creativity to apply AI in problem-solving.

Objective 2: Promote Equity of Access and Participation

Ensure that AI-enabled learning opportunities reach all learners, regardless of socioeconomic background, gender, location (urban vs rural), language or disability. AI should help close existing learning gaps, not widen them. This means targeted interventions for underserved regions, inclusive design of AI tools (e.g., multi-lingual, accessible formats), and monitoring of participation and outcomes to ensure no learner is left behind.

Objective 3: Strengthen Teacher Capacity and Pedagogical Transformation

Support teachers and school leaders to use AI meaningfully, not as mere replacement of teaching, but as an augmentation of pedagogy. This objective focuses on developing teacher competencies in AI-enhanced instructional design, data-driven learner analytics, adaptive feedback systems, and continuous professional development. It also involves institutionalising support structures (mentoring, communities of practice, digital tool libraries) to enable teachers to lead and innovate, rather than simply follow.

Objective 4: Foster Ethical, Responsible and Transparent AI Use

Embed principles of trust, fairness, accountability, human-agency, transparency and data privacy into all AI-in-education initiatives. This means establishing governance frameworks, auditing mechanisms, bias-mitigation protocols, and clear policies for data use and learner protection. The objective ensures that AI is implemented as a human-centred tool that respects rights, preserves dignity, and promotes ethical innovation, rather than risking unintended harms or reinforcing inequities.

Objective 5: Accelerate Innovation, Localisation and Sustainability

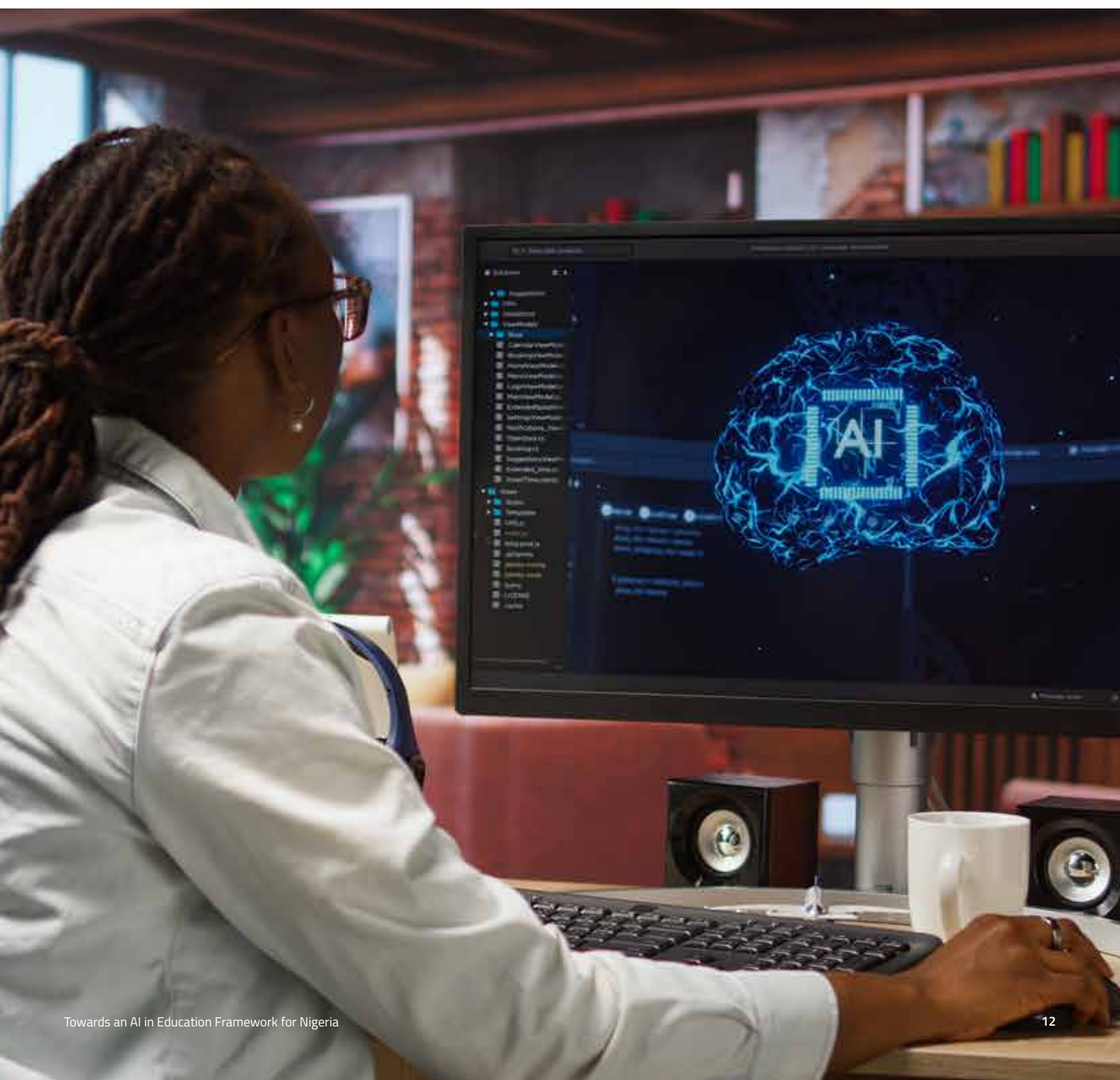
Encourage the creation, adaptation and sustainable scaling of AI solutions that are contextualised to Nigeria's educational realities. This includes supporting local edtech start-ups, open-source platforms, multilingual AI models, partnerships between government, private sector, academia and communities, and long-term funding models. The objective is to move beyond one-off pilots to sustainable ecosystem growth where AI becomes part of the fabric of schooling, teacher professional practice and lifelong learning.

Objective 6: Improve Learning Outcomes, Skills for Future Work and Data-Driven Decision-Making

Leverage AI to enhance learning outcomes (e.g., literacy, numeracy, critical thinking), build 21st-century skills (digital skills, problem-solving, AI-awareness) and support data-driven policy and school-level decision-making.

Box 2: Guiding Principles

Underpinning these strategic objectives are a set of cross-cutting guiding principles including Inclusivity, Equity, Human-Centredness, Transparency, Accountability, Sustainability, and Innovation. These principles ensure that the objectives are pursued not just technically, but ethically and inclusively, in alignment with Nigeria's educational mission and global norms.



2.

Research Methodology

Methodology

This report was developed using a mixed-methods desktop research approach that draws from national policy documents, curriculum frameworks, regulatory guidelines, and recent government announcements on AI education in Nigeria. The methodology also reviewed empirical studies, programme evaluations, and academic literature on digital learning and AI adoption across all levels of education. Global case studies were included to benchmark Nigeria's current progress against emerging best practices.

Stakeholder insights and government communications helped clarify the direction of ongoing reforms, particularly in areas like teacher professional development, curriculum changes, and access infrastructure. This approach allowed for a comprehensive assessment of the current state of AI-in-education readiness and identification of systemic gaps that must be addressed for equitable scale-up.

Theory of Change

The theory of change guiding this work assumes that AI can improve learning outcomes and address long-standing inequities in the education sector if introduced in a structured and ethical manner. The framework links five core enablers: curriculum alignment, teacher capacity, infrastructure and connectivity, inclusive governance, and sustainable financing.

When these elements are strengthened together, AI becomes a tool that empowers teachers, supports personalised learning, enhances skills for future work, and extends educational opportunities to underserved groups including girls, rural learners, and adults. Ultimately, if Nigeria builds a coherent ecosystem that embeds equity and responsible use at the centre of implementation, AI will help accelerate learning and improve human-capital development across the country.

Limitation

The analysis relies primarily on secondary data sources, as a nationwide field assessment of AI use in classrooms has not yet been conducted.

Some policy reforms and pilots, such as the revised curriculum and AI teacher-training programmes, are still at early stages, meaning data on implementation and outcomes remain limited and uneven across states. Infrastructure gaps are particularly under-documented in rural schools and non-formal settings, where access challenges may be more significant than available sources suggest. Rapid technological change also means that some initiatives may evolve faster than published information captures. As AI deployments grow, ongoing field research and updated monitoring will be required to validate assumptions, refine strategies, and ensure that integration remains equitable and consistent with Nigeria's education goals.

Theoretical Framework: Alignment with National Development Priorities

Nigeria's national development agenda emphasizes digital transformation, human-capital strengthening, and global competitiveness. This Framework for AI in Education is therefore not an isolated initiative, but a strategic contribution to broader national goals, ensuring that the education system produces a future-ready workforce capable of thriving in an AI-enabled economy. The National Digital Economy Policy and Strategy (NDEPS 2020–2030) identifies Digital Literacy & Skills, Solid Infrastructure, Digital Society & Emerging Technologies, and Indigenous Content Development as critical pillars. These priorities directly align with this Framework's focus on integrating AI skills into curricula, strengthening connectivity and device access, and elevating local innovation. As the National Information Technology Development Agency (NITDA) underscores, "Our human capital is our greatest resource. If we are serious about reform, we must start with building digital fluency across the population. ... If you're not digitally visible, you risk being excluded." This reinforces the central role of education in enabling equitable participation in Nigeria's digital future.

The Framework also complements policies driving transformation across the education ecosystem. The National Implementation Guidelines for ICT in Education (2019) outlines five levers for digital reform, human capital development, infrastructure, research and innovation,

governance, and monitoring. The government's Human Capital Opportunities for Prosperity and Equity (HOPE) Programme similarly elevates foundational learning, digital skills, and workforce readiness as core to economic renewal. By embedding AI into teaching, learning, and system administration, the Framework enhances coherence across national priorities, while supporting political buy-in and unified implementation across federal, state, and institutional levels.

Finally, this Framework strengthens Nigeria's commitment to equity and inclusive growth. Despite policy progress, persistent disparities limit

digital opportunity for rural schools, girls, and learners with disabilities. National strategies emphasize inclusion as a non-negotiable principle of digital transformation, with NDEPS explicitly highlighting a "Digital Society" that benefits all citizens. AI-enabled education when accompanied by safeguards, infrastructure investments, and gender-responsive design can help close these gaps instead of widening them. In this way, the Framework aligns with Nigeria's broader socio-economic ambitions: improving learning outcomes, expanding opportunity, and ensuring that all learners gain the skills needed to participate fully in the nation's AI-driven development trajectory





Programme and Policy Frameworks on AI Education in Nigeria

Current AI, Education Policies and Laws in Nigeria

The Nigerian Federal Government has publicly signalled its commitment to integrating artificial intelligence within its education and broader digital economy agenda. According to the National Artificial Intelligence Strategy (NAIS) draft, the vision is “to be a global leader in harnessing the transformative power of AI through responsible, ethical and inclusive innovation, fostering sustainable development through collaborative efforts.”¹² The document states: “The advent of Artificial Intelligence (AI) has precipitated a paradigmatic shift ... driving cutting-edge advancements across nations and industries. ... Nigeria, as the continent’s most populous country, is singularly positioned to harness AI’s transformative capacity to address pressing socio-economic conundrums, accelerate economic growth, and pivot the nation into a new epoch of technological empowerment...”¹³ This establishes the strategic importance the government places on AI as a lever for education, innovation and national development.

The strategy outlines five strategic pillars that explicitly reflect the link to sector adoption, including education, and provide the architecture for governance, infrastructure development, ecosystem building and sector transformation. The pillars are listed as:

- i. building foundational AI infrastructure,
- ii. building and sustaining a world-class AI ecosystem,
- iii. accelerating AI adoption and sector transformation,
- iv. ensuring responsible and ethical AI development, and
- v. developing a robust AI governance framework.¹⁴

The pillar of “accelerating AI adoption and sector transformation” points directly to embedding AI in key sectors (including education) and signals the government’s intention to integrate AI beyond just infrastructure and governance into practical application and scale-up in areas such as schooling and skills-development.

Furthermore, the NAIS draft emphasises the need for inclusive and sector-specific application of AI, stating that Nigeria must move beyond “pilot and proof-of-concept” and instead “scale AI solutions across critical sectors such as healthcare, agriculture, education and public services” to deliver socio-economic benefits.¹⁵ For education specifically, this signals a policy environment where AI is not an after-thought but a strategic tool, one that must be governed (pillar five), supported by infrastructure (pillar one), nurtured in ecosystem (pillar two), adopted in sector transformation (pillar three), and developed ethically (pillar four). In doing so, the government is laying the groundwork for AI interventions in Nigeria’s educational systems to be more systemic rather than ad-hoc.

The National Policy on Education (NPE), first promulgated in 1977 and most recently revised in 2013, remains the overarching policy framework guiding all levels of Nigeria’s education system.¹⁶ The document describes the NPE as “a statement of intentions, expectations, goals, prescriptions, standards and requirements for quality education delivery in Nigeria.”¹⁷ Its philosophy emphasises that “education is an instrument for national development and social change,” and that it “maximises the creative potentials and skills of the individual for self-fulfillment and general development of the society.”¹⁸ While the policy does not explicitly mention artificial intelligence, it sets the structural, curricular and institutional foundation needed to integrate modern technologies and innovations. For example, the NPE highlights that teaching shall be “practical, activity-based, experiential and IT supported.”¹⁹

¹² https://ncair.nitda.gov.ng/wp-content/uploads/2024/08/National-AI-Strategy_01082024-copy.pdf

¹³ https://ncair.nitda.gov.ng/wp-content/uploads/2024/08/National-AI-Strategy_01082024-copy.pdf

¹⁴ <https://www.olaniwunajayi.net/blog/wp-content/uploads/2024/09/Overview-of-the-Draft-National-Artificial-Intelligence-Strategy-September-2024-OALP-Tech-Newsletter.pdf>

¹⁵ <https://pavestoneslegal.com/analysis-of-nigerias-national-artificial-intelligence-strategy/>

¹⁶ <https://education.gov.ng/wp-content/uploads/2020/01/lfe-Olatunji-David-Contributor-Panel-3.pdf>

¹⁷ https://rtei.okfn.org/documents/National_Policy_on_Education_2013.html

¹⁸ <https://education.gov.ng/wp-content/uploads/2020/01/lfe-Olatunji-David-Contributor-Panel-3.pdf>

¹⁹ <https://education.gov.ng/wp-content/uploads/2020/01/lfe-Olatunji-David-Contributor-Panel-3.pdf>

The document also outlines the roles of infrastructure (e.g., educational support services), teacher education, curriculum development and administration at all levels, all of which are directly relevant preconditions for the later introduction and scaling of AI-based educational interventions.

The Nigerian Ministry of Education, 2025, mandated the structured integration of artificial intelligence (AI), coding and robotics into the national secondary school curriculum. For example, a ministry communication states: "The meeting concluded with a shared commitment to expanding AI and robotics education in Nigeria, ensuring alignment with national education objectives."²⁰ Further, reporting on the new curriculum reveals that as of September 2025 "the revised curriculum will introduce compulsory modules in artificial intelligence, coding ... starting in the 2025/2026 academic session."²¹ This signal indicates that the government is moving beyond broad policy statements to concrete curriculum interventions embedding emerging tech skills early in schooling rather than treating them as optional or extra-curricular.

A recent study titled "Generative AI Adoption and the Politics of Educational Governance in Nigeria" finds that "persistent policy gaps, digital divides, and governance challenges hinder inclusive adoption."²² The authors emphasise that while AI in education holds promise, the real-world outcomes are mediated by issues of equity: "the role of equity in ensuring that AI tools enhance, rather than deepen, existing disparities between urban and rural learners."²³ Complementing this, a broader regulatory review of AI in Nigeria concludes that "Nigeria lacks a dedicated AI legal framework ... AI-related governance is scattered across data protection, cybersecurity, and competition laws, leading to regulatory fragmentation and enforcement challenges."²⁴ These findings make clear that introducing AI into education must go hand-in-hand with

strengthening ethical standards, data governance, teacher capacity, and access infrastructure, otherwise the gains risk being uneven or inequitable.

3.2

Current Education Programmes on AI Education in Nigeria.

Nigeria is already witnessing a growing set of programmes aimed at integrating artificial intelligence (AI) and related digital technologies into its education ecosystem. One of the most prominent is the pilot conducted in Edo State, which deployed generative AI for an after-school tutoring intervention in senior secondary schools. Over six weeks, students used AI-driven tools (e.g., generative conversational agents) to supplement their regular instruction, and the results showed meaningful learning gains.²⁵ This demonstrates a concrete, scaled pilot of AI in schooling contexts and positions Nigeria among early adopters of AI-driven tutoring in Sub-Saharan Africa.

Another initiative is by the Nigerian government itself: the 3 Million Technical Talent (3MTT) Programme launched in late 2023 aims to train three million Nigerians in high-demand digital skills including artificial intelligence, machine learning and data science by 2027.²⁶ Although not exclusively an "education in schools" programme, it is a major national initiative oriented towards building capacity in AI and related domains and therefore directly relevant to the education-ecosystem readiness.

On the private-sector front, the Nigerian edtech company Bildup AI has launched an ambitious "AI in Every Classroom" initiative, under which it is investing roughly NGN 500 million to equip schools with AI-powered personalized learning tools that adapt to each pupil's level and pace.²⁷ This represents another strand of AI-education

²⁰ <https://fmino.gov.ng/ai-coding-and-robotics-minister-of-education-calls-for-structured-integration-in-nigerias-curriculum/>

²¹ <https://businessday.ng/education/article/new-secondary-school-curriculum-to-include-ai-coding-journalism-from-september-2025/>

²² <https://iiardjournals.org/get/WJIMT/VOL.%209%20NO.%208%202025/GENERATIVE%20AI%20ADOPTION%2099-111.pdf>

²³ <https://iiardjournals.org/get/WJIMT/VOL.%209%20NO.%208%202025/GENERATIVE%20AI%20ADOPTION%2099-111.pdf>

²⁴ https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5117653&

²⁵ <https://blogs.worldbank.org/en/developmenttalk/addressing-the-learning-crisis-with-generative-ai--lessons-from->

²⁶ <https://en.wikipedia.org/wiki/3MTT>

²⁷ <https://iafrica.com/bildup-ai-launches-ngn500m-ai-in-every-classroom-initiative-in-nigeria/>

programming, driven by private innovation and school-level deployment.

Additionally, the government introduced teacher training initiatives: for example, the National Senior Secondary Education Commission (NSSEC) announced a nationwide AI training for 6,000 senior secondary-school teachers to enhance teaching methods and prepare them to integrate AI into classrooms.²⁸ These capacity-building efforts are important because teacher readiness is a key determinant of effective AI adoption.

3.3

International Instruments on Inclusive AI in Education

In developing this Framework, it is critical to align with global norms and instruments around inclusive, responsible AI in education. Key references include:

Recommendation on the Ethics of Artificial Intelligence (UNESCO, 2021): This Recommendation, adopted by UNESCO's 193 Member States on 23 November 2021, is the first global normative instrument dedicated to AI ethics.²⁹ It explains clearly that "the protection of human rights and dignity is the cornerstone of the Recommendation, based on the advancement of fundamental principles such as transparency and fairness, always remembering the importance of human oversight of AI systems."³⁰ The document sets out both foundational values (such as respect for human rights, human dignity, inclusiveness, diversity) and policy action areas (including data governance, environment & ecosystems, gender, education and research).³¹ While it is not uniquely focused on education, it explicitly includes "education and research" as one of its eleven policy action areas.³²

For an AI-in-education framework in Nigeria, this instrument signals a strong imperative to integrate ethics, equity, and human-centred oversight into every stage of AI deployment in schooling. It

means that when AI-driven tools are introduced in classrooms, the design, deployment and monitoring must ensure fairness (e.g., no bias against girls or rural students), transparency (students and teachers understand how the AI works), accountability (who is responsible for outcomes) and human-centred oversight (teachers remain essential). Because the Recommendation emphasises human dignity and rights, any African-context adaptation (such as Nigeria's) should embed contextual safeguards: data-privacy protections, inclusive access (including marginalised learners), capacity-building for teachers and learners, and mechanisms for audit and accountability aligned with this global standard.

Guidance for Generative AI in Education and Research (UNESCO, 2023): Published in 2023, UNESCO's Guidance for Generative AI in Education and Research provides global policy direction specifically for how generative artificial intelligence (GenAI) should be approached in both education and research settings.³³ The document states that "Building on UNESCO's Recommendation on the Ethics of Artificial Intelligence ... the guidance is anchored in a human-centred approach that promotes human agency, transparency and public accountability."³⁴ It includes an overview of how GenAI works, identifies major ethical and pedagogical concerns (such as data-privacy, bias, ownership of output, and oversight), and outlines concrete policy and regulatory steps for example mandating data-protection frameworks, considering age-limits for GenAI use in schooling, and validating tools for pedagogical appropriateness.³⁵

The guidance emphasises that GenAI must be managed not just as a technological add-on, but through governance, capacity-building and ethical design.³⁶ It also highlights the need for inclusive access (covering linguistic diversity, learners with disabilities) and the importance of long-term reflection on how GenAI might shift teaching, learning and assessment practices.

²⁸ <https://www.premiumtimesng.com/news/top-news/787060-expert-advocates-ai-driven-curriculum-to-transform-nigerias-education-system.html>

²⁹ <https://www.ohchr.org/sites/default/files/2022-03/UNESCO.pdf>

³⁰ <https://digitallibrary.un.org/record/4062376?ln=en&>

³¹ <https://www.unesco.org/en/articles/recommendation-ethics-artificial-intelligence>

³² https://es.wikipedia.org/wiki/Recomendaci%C3%B3n_sobre_la_%C3%A9tica_de_la_inteligencia_artificial_de_la_UNESCO

³³ <https://unesdoc.unesco.org/ark:/3A/48223/pf0000386693>

³⁴ <https://unesdoc.unesco.org/ark:/3A/48223/pf0000386693>

³⁵ <https://unesdoc.unesco.org/ark:/3A/48223/pf0000386693>

³⁶ <https://www.unesco.org/en/articles/guidance-generative-ai-education-and-research>

Principles for the Ethical Use of Artificial Intelligence in the United Nations System (UN Systemwide, circa 2019-2020):

The United Nations system adopted this set of ten guiding principles to steer the design, development, deployment and oversight of artificial intelligence across its entities. According to the document, the principles “provide a basis for a framework for the ethical use of artificial intelligence (AI) by United Nations system organizations throughout all stages of an AI system lifecycle.”³⁷ They include key concepts such as *“do no harm; defined purpose, necessity and proportionality; safety and security; fairness and non-discrimination; sustainability; right to privacy, data protection and data governance; human autonomy and oversight; transparency and explainability; responsibility and accountability; and inclusion and participation.”*³⁸ The document situates these principles within the UN Charter and human rights law, noting that “an ‘ethical approach’ to the use of artificial intelligence ... is one that a) is consistent with the Charter of the United Nations, respecting all applicable rules of international human rights law, including the right to privacy, as well as with the principles identified below, and b) that entails the assessment of such consistency at all stages of the AI lifecycle.”³⁹

This normative instrument is especially helpful because it covers the entire lifecycle of AI systems from research and design through to deployment, operation, maintenance, monitoring and de-commissioning.⁴⁰ For example, one of the principles states that “AI systems should not be used in ways that cause or exacerbate harm... and the intended and unintended impact of AI systems, at any stage in their lifecycle, should be monitored in order to avoid causing or contributing to harm.”⁴¹ Another emphasises the need for human autonomy and oversight: “The United Nations system organizations should ensure that AI systems do not overrule freedom and autonomy of human beings and should guarantee human oversight.”⁴² These provisions make it clear that

any educational-AI framework needs to embed robust safeguards, ongoing monitoring and mechanisms to protect rights and ensure meaningful human-machine relationships.

Beijing Consensus on Artificial Intelligence and Education (UNESCO, 2019):

The Beijing Consensus, endorsed at the International Conference on Artificial Intelligence and Education in Beijing, China (16–18 May 2019), marks the first major global policy document by UNESCO dedicated to the integration of artificial intelligence into education systems.⁴³ It sets out a vision grounded in the belief that AI-driven innovation in education can play a pivotal role in advancing the SDG-4 agenda (ensuring inclusive and equitable quality education and promoting lifelong learning). For instance, the document affirms that the deployment of AI technologies “should be in the service of people to enhance human capacities ... and that the impact of AI on people and society should be monitored and evaluated throughout the value chains.”⁴⁴ Additionally, it emphasises that such planning must follow “whole-of-government, inter-sectoral and multi-stakeholder approaches”, reflecting the complexity of embedding AI into educational systems.⁴⁵

Among its recommendations, the Consensus urges policy-makers to integrate AI into education policy frameworks in ways that support lifelong and flexible learning, teacher empowerment, equitable access and ethical data use.⁴⁶ For example, it states that stakeholders should “ensure that AI technologies are used to empower teachers rather than replace them,” and should promote “equitable and inclusive use of artificial intelligence in education ... irrespective of gender, disability, social or economic status, ethnic or cultural background or geographic location.”⁴⁷ In doing so, the document places human agency, inclusion, and equity at the heart of the AI-in-education discourse.

³⁷ <https://unsceb.org/principles-ethical-use-artificial-intelligence-united-nations-system>

³⁸ <https://unsceb.org/principles-ethical-use-artificial-intelligence-united-nations-system>

³⁹ <https://www.ictworks.org/artificial-intelligence-principles-united-nations/>

⁴⁰ <https://unsceb.org/principles-ethical-use-artificial-intelligence-united-nations-system>

⁴¹ <https://www.ictworks.org/artificial-intelligence-principles-united-nations/>

⁴² <https://www.diplomacy.edu/blog/ethics-and-ai-part-4/>

⁴³ <https://www.unesco.org/en/articles/first-ever-consensus-artificial-intelligence-and-education-published-unesco>

⁴⁴ https://www.researchgate.net/publication/386238469_Beijing_consensus_on_artificial_intelligence_and_education

⁴⁵ <https://www.unesco.org/en/articles/first-ever-consensus-artificial-intelligence-and-education-published-unesco>

⁴⁶ <https://unesdoc.unesco.org/ark:/48223/pf0000368303>

⁴⁷ https://en.moe.gov.cn/news/press_releases/201909/t20190902_396913.html

Organisation for Economic Co operation and Development (OECD) - Recommendation on Artificial Intelligence & related education

guidance: The OECD Recommendation on Artificial Intelligence provides the first intergovernmental standard on AI, adopted by Member States in May 2019, and updated in May 2024.⁴⁸ The accompanying “AI Principles” emphasise that AI should be “innovative and trustworthy and ... respect human rights and democratic values.”⁴⁹ Key values listed include: *inclusive growth, sustainable development and well-being; human rights and democratic values (including fairness and privacy); transparency and explainability; robustness, security and safety; accountability.*⁵⁰ The Recommendation also outlines actionable policy recommendations for governments, such as investing in AI research and development, fostering inclusive ecosystems, shaping interoperable governance, building human capacity, and promoting international cooperation.⁵¹ In the education domain, the OECD recognises that as AI rapidly advances, education systems must assess how AI may automate or complement human tasks, affect learning and teaching, and shift skills demand.

International Telecommunication Union (ITU) / UNESCO – AI Skills Coalition & Working Groups for AI & Education: The ITU, in collaboration with UNESCO and other partners, has launched the “AI Skills Coalition” described as “a UN-led global, open, trusted and inclusive platform ... Priority is given to learners from developing and Least Developed Countries, with a strong emphasis on women and girls.” (AI for Good: AI Skills Coalition).⁵² Additionally, ITU Secretary-General Doreen Bogdan Martin remarked that “those 2.6 billion risk missing out altogether on this AI transformation revolution. We can’t let that happen.”⁵³ These initiatives are explicitly designed to bridge global divides in AI access, focusing on upskilling and reskilling individuals in developing contexts, especially young people, women and

girls so they can leverage AI rather than be left behind.

The Coalition emphasises capacity-building, inclusive access and the necessity of global collaboration to scale AI education. For example, the ITU press release states the platform will offer “free resources such as self-paced courses, webinars, in-person workshops, hybrid programmes, and a comprehensive digital library of AI materials.”⁵⁴ In this way, the initiative goes beyond rhetoric: it aims to provide tangible pathways for learners in underserved regions to gain AI-related competencies. The fact that it is UN-led, global and prioritises developing and least developed countries lends it normative weight—meaning education-sector frameworks (such as for Nigeria) should reflect these capacity and inclusion priorities when integrating AI into learning systems.

Sustainable Development Goal 4 (SDG 4) – Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all;

SDG 4 sets the global vision for education, committing by 2030 to “ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.”⁵⁵ The goal is broken down into multiple targets that map across the education ecosystem, including Target 4.1 “By 2030, ensure that all girls and boys complete free, equitable and quality primary and secondary education leading to relevant and effective learning outcomes.”⁵⁶ Target 4.4 calls for substantially increasing “the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship.”⁵⁷ And Target 4.a emphasises the need to “build and upgrade education facilities that are child, disability and gender sensitive and provide safe, non-violent, inclusive and effective learning environments for all.”⁵⁸ These targets highlight not only access, but also the quality, relevant skills, and inclusive

⁴⁸ <https://oecd.ai/en/ai-principles>

⁴⁹ <https://oecd.ai/en/ai-principles>

⁵⁰ <https://oecd.ai/en/dashboards/policy-initiatives/oecd-ai-principles-9705>

⁵¹ <https://www.ansi.org/standards-news/all-news/5-9-24-oecd-updates-ai-principles>

⁵² <https://aiforgood.itu.int/ai-skills-coalition/>

⁵³ <https://www.itu.int/hub/2025/05/the-ai-skills-coalition-democratizing-ai-education/>

⁵⁴ <https://cadeproject.org/updates/itu-launches-global-ai-skills-coalition-to-bridge-expertise-gap-in-developing-nations/>

⁵⁵ <https://sdgs.un.org/goals/goal4>

⁵⁶ <https://globalgoals.org/goals/4-quality-education/>

⁵⁷ <https://ourworldindata.org/sdgs/quality-education>

⁵⁸ https://www.right-to-education.org/sites/right-to-education.org/files/resource-attachments/A4ID_SDG%20Legal%20Guide_SDG4_Quality%20Education_February2022_EN.pdf

infrastructure dimensions of education.

Beyond laying out targets, SDG 4 underscores the imperative that education systems must leave no one behind: it stresses addressing disparities such as gender, disability, rural/urban and wealth - for example Target 4.5 mandates eliminating gender disparities and ensuring equal access for "persons with disabilities, indigenous peoples and children in vulnerable situations."⁵⁹ The monitoring frameworks note that while enrolments have

increased, quality remains a major challenge, noting that in many regions children are in school but not acquiring basic literacy and numeracy skills.⁶⁰ For a policy framework that incorporates AI in education, SDG 4 serves as a normative anchor indicating that any AI-enabled initiative must advance equity, build relevant skills, improve learning outcomes, and strengthen inclusive environments rather than merely deploying technology for its own sake.

⁶⁰ <https://datatopics.worldbank.org/sdgatlas/goal-4-quality-education/>





Nigeria's Education Curriculum and AI Integration

In Nigeria, while the broader education policy frameworks reference ICT integration and digital skills, there is currently limited systematic incorporation of AI-specific concepts, skills and ethical frameworks into national curricula. For instance, the recently published revised curriculum

for junior and senior secondary schools (effective September 2025) introduces subjects such as Digital Literacy & Coding at the Junior Secondary level and Programming, Artificial Intelligence, Data Science, Cybersecurity at the Senior Secondary level.⁶¹

TABLE 1 Structure of the New Curriculum (2025 Edition)

Level	Number of Subjects (New)	Key Changes and Subjects
Primary 1–3	9–10 subjects	English Studies, Mathematics, One Nigerian Language (Hausa, Igbo, or Yoruba), Basic Science, Physical & Health Education, Christian/Islamic Religious Studies, Nigerian History, Social & Citizenship Studies, Cultural & Creative Arts; Arabic optional
Primary 4–6	11–13 subjects	Includes Basic Science & Technology, Digital Literacy, Nigerian History (compulsory), Citizenship Studies, and French optional
Junior Secondary (JSS 1–3)	12–14 subjects	Includes Intermediate Science, Digital Technologies, Nigerian History, Pre-vocational Studies or Trade subjects
Senior Secondary School (SS 1–3)	8–9 subjects	Five compulsory subjects: English Language, General Mathematics, Citizenship and Heritage Studies, Digital Technology, One Trade/Entrepreneurship subject (e.g. Solar Installation, Computer Hardware and GSM Repairs, Fashion Design, etc.). Field of study Sciences, Humanities, and Business, etc.

Source: FMOE, September 2025⁶². *Note:* The Federal Ministry of Education announced that the new national curriculum will take effect in the 2025/2026 academic session, following joint work with NERDC, NECO, WAEC, and other agencies. The updated subject structure reduces the number of core courses per level for Primary 1–6:

However, despite these curricular additions, deeper AI-learning elements, such as machine learning, data-literacy, computational thinking, and AI ethics, remain rare, and the rollout is still uneven across primary, secondary and tertiary levels. A recent review notes that although Nigeria has launched initiatives such as the National Information Technology Development Agency (NITDA)-led Digital Literacy and Skills Curriculum covering emerging technologies such as AI at tertiary level, the actual classroom integration of AI-driven pedagogy and content remains nascent.⁶³

Several specific gaps persist:

- 1. Teacher and institutional capacity,** many teachers lack preparation for delivering AI curriculum modules, and many schools lack the infrastructure to support AI-enabled learning. A study on digital literacy in Nigerian tertiary institutions found that although AI, machine learning and data science are emerging trends, the major barrier was limited access to technology and skill-sets.⁶⁴

⁶¹ <https://guardian.ng/features/education/full-list-subjects-for-jss-sss-as-fg-rolls-out-new-national-curriculum/>

⁶² <https://education.gov.ng/wp-content/uploads/2025/09/FG-OVERHAULS-CURRICULUM.pdf>

⁶³ <https://www.itedgenews.africa/nitda-nuc-to-fast-track-digital-literacy-curriculum-implementation-in-nigerian-universities/>

⁶⁴ https://www.irjms.com/wp-content/uploads/2025/04/Manuscript_IRJMS_02243_W5.pdf

2. **Learner progression and module coherence**, while primary levels may include basic digital literacy instruction, there is little coherent, age-appropriate progression into full AI literacy in lower and upper secondary levels. For example, although the primary school curriculum introduces basic digital literacy for young children, articles note that “modules on AI fundamentals, machine-learning, data-literacy or computational thinking remain rare and unevenly distributed”.⁶⁵
3. **Equity and distribution issues**, the rollout of new digital or AI-inflected curriculum content runs the risk of reinforcing existing disparities (urban/rural, resource-rich vs under-resourced schools) since infrastructure and teacher readiness are highly variable.
4. **Governance, ethics and content localisation**, the focus has thus far been more on digital literacy broadly rather than on embedding AI ethics, bias-awareness, algorithmic understanding, or culturally localized AI content in curricula. For example, the national digital policy emphasizes digital literacy but less so explicit AI ethics in school curricula.

Global Examples of AI Education Curriculum (Case Study / Best Practices): Several global models illustrate how education systems are incorporating AI concepts, skills and ethics into curricula. For example, the “Day of AI” curriculum (developed by MIT RAISE and partners) provides a tiered, spiral-learning approach from early ages (5–7 years) through older learners, with modules such as “What is AI?”, “How Do Machines Learn?”, “Bias in AI”, “Generative AI and Deep Fakes”.⁶⁶ In the U.S., the U.S. Department of Education released a policy brief on “Artificial Intelligence and the Future of Teaching and Learning” which emphasises the need for AI-aware pedagogy, teacher professional development, and ethical frameworks for system-wide integration.⁶⁷

From the literature: a study reviewing AI curricula in eight countries (United States, Finland, Australia, Singapore, India, China, South Korea, UK) observed rising adoption of national guidelines for K-12 AI education, though many models still emphasise computational thinking and coding, with less systematic coverage of ethics and cross-disciplinary applications.⁶⁸

These examples offer useful templates for Nigeria’s adaptation: clear pathways by age/grade, integration of AI fundamentals, ethics, data-literacy, and cross-disciplinary linkages spanning STEM, humanities and vocational tracks.

Curriculum Integration Framework: Key Elements

To ensure effective embedding of AI concepts, skills and ethics into Nigeria’s educational system, the following key elements form the core of the curriculum integration framework:

1. **Systematic Incorporation of AI Concepts, Skills, and Ethics:** Rather than treating AI as a standalone elective, the curriculum must integrate foundational AI concepts (e.g., what is AI, machine-learning basics), data-literacy (e.g., interpreting data, recognizing bias), computational thinking (e.g., algorithms, logical reasoning) and ethical frameworks (e.g., fairness, transparency, human-agency) across learning levels. As noted in a recent UNESCO student competency framework, key elements include: “*A human-centred mindset; Ethics of AI; AI techniques and applications; AI system design.*”⁶⁹
2. **Tailored Learning Outcomes and Proficiency Benchmarks by Education Level:** The framework should define progressive proficiency levels aligned to primary, secondary and tertiary (or vocational) stages. For example, younger learners may focus on “recognise AI in daily life,” moving to “apply simple AI tools and

⁶⁵ <https://thenigeriaeducationnews.com/2025/09/03/nigerias-new-primary-school-curriculum-a-comprehensive-breakdown-of-learning-from-primary-1-to-senior-secondary/>

⁶⁶ <https://dayofai.org/curriculum/>

⁶⁷ <https://www.ed.gov/sites/ed/files/documents/ai-report/ai-report.pdf>

⁶⁸ <https://icce2021.apsce.net/wp-content/uploads/2021/12/ICCE2021-Vol.II-PP-769-773.pdf>

⁶⁹ <https://www.unesco.org/en/articles/what-you-need-know-about-unescos-new-ai-competency-frameworks-students-and-teachers>

evaluate results,” and ultimately “design and ethically evaluate AI systems.” This nuance is emphasised in international policy advice, such as by the World Economic Forum which notes: “Students and educators must learn about AI ... including understanding AI’s principles, applications and ethical implications.”⁷⁰

3. **Modules Covering AI Fundamentals, Machine Learning, Data-Literacy & Computational Thinking:** A well-rounded AI curriculum includes modules such as algorithms and logic, data collection and cleaning, supervised/unsupervised learning concepts, human-computer interaction, and ethical issues (e.g., bias, data privacy). Guidance for K-12 integration emphasises that “by embedding AI concepts into various subjects, students can develop essential digital-age skills across learning environments: school, home and community.”⁷¹
4. **Cross-Disciplinary Linkages: STEM, Social Sciences, Humanities, Vocational Courses:** AI should not be siloed within computer science; instead, it should connect to other domains. For instance: social science classes might discuss AI’s societal impact, humanities could explore AI ethics and culture, and vocational/technical courses could apply AI to agriculture, health-tech or manufacturing. Such interdisciplinarity aligns with the human-centred and inclusive approach that UNESCO advocates in its AI competency frameworks.⁷²

5. **Localization, Multilingualism and Contextual Relevance:** The curriculum must reflect Nigeria’s linguistic diversity, cultural context and varied schooling environments. While international sources may not always explicitly state this, the underlying principles of inclusive, equitable AI literacy imply adaptation to local languages, contexts and values.
6. **Continuous Professional Development for Teachers and Infrastructure Enablement:** Teachers must be equipped not only to deliver content but to facilitate inquiry-based, project-based AI learning and to guide ethical discussions. Simultaneously, infrastructure (devices, connectivity, power) must support AI modules. The U.S. Department of Education’s report emphasises the need to “inspect, explain, override AI” and ensure teachers stay central rather than being replaced.⁷³
7. **Monitoring, Evaluation & Iterative Refinement:** The curriculum framework must embed mechanisms for tracking learner outcomes, access, equity (gender, rural/urban), and tool use. Feedback loops should inform revisions and scaling. International policy guidance emphasises aligning AI tools to a shared vision for learning, ensuring trust and safeguards.⁷⁴

⁷⁰ <https://www.weforum.org/stories/2024/04/prepare-future-policy-ideas-ai-in-education/>

⁷¹ <https://dpi.wi.gov/imt/ai-guidance/educators/curriculum-integration>

⁷² <https://www.unesco.org/en/articles/what-you-need-know-about-unescos-new-ai-competency-frameworks-students-and-teachers>

⁷³ <https://www.ed.gov/sites/ed/files/documents/ai-report/ai-report.pdf>

⁷⁴ <https://www.ed.gov/sites/ed/files/documents/ai-report/ai-report.pdf>

5.

AI Education Across Different Learning Levels

This section outlines how AI in education can be systematically tailored to learners' developmental stages from early childhood through adult education and emphasises inclusion, safe use and lifelong learning opportunities for marginalised groups.

5.1

Early Childhood & Primary Education

Current Situation / Gaps: Global evidence shows that most AI literacy programs are designed for secondary or higher education, often overlooking primary-level learners. A 2023–2025 scoping review across early childhood studies found that teachers rarely introduce AI beyond basic digital or computational thinking, and explicit lessons on algorithms or data ethics are “largely absent” in early years classrooms. Studies by Yim (2024) and Su et al. (2023) also underline that curriculum designers remain wary of exposing very young children to technical AI topics, even though creative, play-based and visual approaches to AI awareness have shown strong potential outcomes.⁷⁵

In Nigeria, primary education still focuses on digital literacy and ICT competence, not AI literacy.

Practical experiences like the character Ayomide learning that AI cannot replace human creativity highlight how AI literacy can be introduced using relatable narratives for younger learners, making abstract concepts easier to grasp through familiar school and home experiences.⁷⁶

A recent study found that AI literacy education “mainly targets secondary and university students, often overlooking the unique needs of younger students.”⁷⁷ While digital literacy may be part of curricula in many Nigerian primary schools, modules that explicitly address AI concepts (e.g., pattern recognition, data vs algorithm, machine-learning basics) are rarely present. A 2025 analysis by TechCabal Insights noted that only 5.67% of public primary schools have computer facilities, compared to 54.55% of private schools, creating major structural inequality.⁷⁸ While the Federal Government’s National AI Policy (2023) and its training program “3MTT” seek to improve readiness, these policies currently emphasize workforce and tertiary learning, not early childhood education. A 2025 article by Mary Eyajuwa Adole emphasized that AI could enhance literacy acquisition and engagement in classrooms, but implementation depends on teacher training and infrastructure readiness.⁷⁹

Many primary schools, especially in rural or underserved areas, lack reliable connectivity, appropriate devices or teacher training in AI-oriented pedagogy. African and Nigerian teachers face steep challenges in adopting AI tools in basic education. A continental review titled AI in African Schools (2025) identified the lack of teacher training, inadequate professional digital culture, and transmissive-style curricula as central barriers. Even where technologies exist, teachers often lack both methodological support and policy integration frameworks to effectively apply AI pedagogies.⁸⁰ Research from UNESCO and Dalberg (2024–2025) confirms similar findings: that without targeted teacher reskilling, AI education strategies will remain largely aspirational.⁸¹

Global Case Study / Best Practice: International examples demonstrate how early-childhood and primary schooling can incorporate AI literacy in age-appropriate and engaging ways. For instance, a study on AI education for young children

⁷⁵ <https://files.eric.ed.gov/fulltext/EJ1470438.pdf>

⁷⁶ <https://drive.google.com/file/d/1qGjrtkp1Z2n25A2i8XNBCpZUqw9L7xW1/view>

⁷⁷ <https://www.sciencedirect.com/science/article/pii/S2666920X2400122X>

⁷⁸ <https://insights.techcabal.com/strengthening-nigerias-ai-literacy-drive-through-real-reform/>

⁷⁹ <https://scholarj.com/index.php/education/article/download/105/100>

⁸⁰ <https://www.unesco.org/ethics-ai/en/articles/ai-literacy-and-new-digital-divide-global-call-action>

⁸¹ <https://dalberg.com/our-ideas/transforming-classrooms-the-growing-impact-of-artificial-intelligence-in-african-education/>

⁸² <https://onlinelibrary.wiley.com/doi/10.1111/jcal.12892>

implemented a six-week programme in a kindergarten setting using intelligent agents and found positive engagement and conceptual understanding among learners.⁸² Another design-case study documents how educators and administrators co-designed an AI literacy curriculum for elementary students using a design-thinking approach, emphasising human-agent interaction and scaffolded understanding of AI basics.⁸³ These cases suggest that Nigeria's framework could benefit from adapting approaches such as: introducing AI concepts via interactive, playful learning; building teacher capacity in early pedagogy of AI; and providing scaffolded progression from awareness to deeper understanding all within the resource constraints of primary education.

Research on K-12 AI education shows that activities such as game-based learning, unplugged computational thinking, and visual modelling help younger learners begin building AI awareness. For example, a study on "AI learning tools in K-12 education" outlines curricula designed for primary and secondary learners, with modules in "What is AI?", "Data and Decisions", "Bias in Systems".⁸⁴ One pilot ("A Structured Unplugged Approach for Foundational AI Literacy in Primary Education") found significant gains in children's ability to describe AI decision processes and reason about data.⁸⁵ These models show how Nigeria's framework could include age-appropriate AI exposure (e.g., in upper primary) to build foundation for secondary and beyond.

on primary and secondary contexts.⁸⁶ In Nigeria, this gap is compounded by unequal digital infrastructure, variable teacher preparedness and limited curriculum modules focused on AI, data-literacy or computational thinking. Moreover, cross-disciplinary AI linkages (in STEM, social sciences, humanities) are rarely institutionalised, and the ethics of AI (bias, data-privacy) are seldom addressed in schools.

Studies on AI in secondary education in Nigeria reveal both promising potential and substantial gaps. A randomized controlled trial in World Bank-supported pilot in Edo State found that first-year senior secondary students using a generative AI tutor improved by 0.31 standard deviations, equivalent to roughly 1.5–2 years' worth of typical schooling gains, with the largest effects for female students.⁸⁷ Meanwhile, a survey of 174 secondary school teachers in Lagos found that although most are aware of AI software, "the majority ... do not know how it works" and awareness of AI concepts among pre-service teachers is particularly weak.⁸⁸ Other research into Nigeria's secondary school system highlights that AI adoption is still "in its infancy," hampered by infrastructure deficits, data-privacy issues, resistance among educators and insufficient institutional capacity

Global Case Study / Best Practice: International experience demonstrates that successful AI integration in secondary schools combines strong conceptual grounding with hands-on, ethical, and interdisciplinary learning.⁸⁹ Countries such as China, the United States, and EU member states have introduced curriculum pathways where learners progress from foundational knowledge to model-building and real-world applications. For instance, China's General AI Education Guide (2025) scaffolds students from introductory logic and core principles in lower grades to basic machine-learning and algorithm design at senior secondary level^{90,91,92} while U.S. and European

Secondary Education

Current Situation / Gaps: At the secondary level, there is greater potential for integrating AI concepts, but adoption remains patchy in many systems. A systematic review of AI in education found that higher education dominates the research landscape and that fewer studies focus

⁸³ https://edtechbooks.org/jaid_14_3/nsuqflljjsd

⁸⁴ <https://link.springer.com/article/10.1007/s40692-023-00304-9>

⁸⁵ <https://arxiv.org/abs/2505.21398>

⁸⁶ <https://www.mdpi.com/2414-4088/9/8/84>

⁸⁷ <https://documents1.worldbank.org/curated/en/099548105192529324/pdf/IDU-c09f40d8-9ff8-42dc-b315-591157499be7.pdf>

⁸⁸ <https://educationalperspectives.org.ng/read.php?id=261&>

⁸⁹ <https://www.clarity-innovations.com/portfolio/international-society-technology-education/high-school-project-based-learning-unit-about>

⁹⁰ http://english.scio.gov.cn/pressroom/2025-05/13/content_117871666.html

⁹¹ <https://www.education.gov.au/download/19495/chinas-approach-ai-education-primary-and-secondary-schools/41945/document/pdf>

⁹² <https://theaitrack.com/china-mandates-ai-education/>

frameworks increasingly include core topics such as neural networks, data science workflows, supervised versus unsupervised learning, and human–AI interaction.^{93,94} Some K-12 programs in China and elsewhere are teaching modules on machine-learning, AI project design and ethical implications.⁹⁵ These provide strong templates for Nigeria’s secondary curriculum: e.g., project-based modules where students design simple AI applications, analyse data, reflect on bias/ethics and link across disciplines. These approaches ensure students build both cognitive understanding and technical fluency.

A defining feature of global best practice is project-based learning: students design simple AI systems chatbots, recommendation engines, classification tools, or carry out collaborative data-analysis projects that require applying Python, Scratch, or cloud-based AI tools.⁹⁶ Such activities foster real-world problem solving and situate AI in authentic contexts, from environmental sustainability to health-care simulations to artistic expression.⁹⁷ The intentional cross-disciplinary design helps learners connect AI to social studies, digital ethics, and civic participation, strengthening critical thinking and cultural relevance.⁹⁸

TABLE 2 International Secondary AI Curriculum Features

Feature	Example Programs/Countries	Details
Machine learning basics	China, US, EU	Core principles in simple, real projects
Project-based modules	Global (PBL, AI4K12, ISTE, China)	Student-led AI app design, cross-curricular
Data analysis and coding	US, EU, China	Python, Scratch, hands-on data projects
Ethics, bias, privacy	China, OECD, US, UNESCO	Reflection modules, debates, authentic tasks
Cross-disciplinary linkages	EU, China, Global Ed. Bodies	AI in environmental science, humanities, etc.
Tiered, age-appropriate design	China, OECD, US	Matched to student cognitive stage
Teacher training	China, PBL initiatives	Toolkits, workshops, support

Source: UNESCO 2022, Xinhua 2025, ISTE 2025, and Bosarge 2025.

Ethical and responsible AI use is embedded by design. International curricula include structured discussions and case studies on algorithmic bias, data-protection rules, misinformation, and the risks of over-automating judgement. Many systems explicitly prohibit students from submitting unedited AI-generated work in assessments, teaching learners to critically evaluate AI outputs rather than bypass engagement. These initiatives invest heavily in teacher readiness. Tiered instructional models

align curriculum complexity to student development,⁹⁹ while continuous professional development and classroom toolkits ensure educators, regardless of prior technical background, can confidently guide learning. These global lessons provide a strong template for Nigeria as it seeks to deepen AI integration at the secondary level: focusing not only on content adoption, but on pedagogy, ethics, teacher competence, and meaningful learner engagement.

⁹³ <https://futureofbeinghuman.com/api/v1/file/2737ea4a-2ca4-4cb5-85cb-4c9ad88f6204.pdf>

⁹⁴ <https://critical-links.com/wp-content/uploads/2025/04/K-12-AI-Curricula-UNESCO.pdf>

⁹⁵ <https://link.springer.com/article/10.1007/s40692-023-00304-9>

⁹⁶ https://cdn.iste.org/www-root/Libraries/Documents%20&%20Files/Artificial%20Intelligence/AIGDSE_1120.pdf

⁹⁷ <https://aixeducation.substack.com/p/project-based-learning-with-ai>

⁹⁸ <https://www.education.gov.au/download/19495/chinas-approach-ai-education-primary-and-secondary-schools/41945/document/pdf>

⁹⁹ http://english.scio.gov.cn/pressroom/2025-05/13/content_117871666.html

Box 3: Lesson from World Bank Assisted AI in Secondary Education Model in Edo State

Nigeria's first randomized evaluation of AI tutoring in secondary schools finds large, rapid learning gains that extend beyond the target subject. Over six weeks, SS1 students using Microsoft Copilot (GPT-4) as a virtual English tutor improved by 0.31 σ on composite assessments spanning English, AI knowledge and digital skills; the English-only effect was 0.23 σ , which the authors benchmark against roughly 1.5–2 years of typical schooling, with clear positive spillovers to digital literacy and AI awareness. Effects scale with participation: every extra attendance day added 0.031 σ , and simple projections suggest a full academic year of consistent use could yield 1.2–2.2 σ gains. Results remain robust across specifications and are not driven by any single school, with conservative controls for baseline performance and attrition.

Cost metrics position the intervention among the most efficient learning investments in low-resource settings. Per-student costs were \$48 in the pilot (marginal \$9), with a projected \$124 for a full-year program. When compared to global benchmarks, the study estimates benefit-cost ratios up to 260:1 and learning-adjusted years of schooling (LAYS) up to 0.9 per participant. Critically, these impacts were achieved despite frequent real-world frictions, connectivity drops, power outages, and device constraints, demonstrating that AI tutoring can produce outsized gains even in challenging delivery environments when paired with light-touch teacher facilitation and aligned content.¹⁰⁰

Equity results are encouraging but uneven, underscoring design choices for scale-up. All subgroups benefited, with the largest gains for girls, who began behind peers followed by higher prior performers and students from relatively better-off households; learners from lower-income households improved too, but by less, likely reflecting lower baseline digital familiarity. The policy read-through is clear: mainstreaming AI tutoring will require teacher upskilling, curriculum alignment and strong norms to discourage shortcutting, alongside investments that narrow the digital-access gap (infrastructure, devices, safe spaces) so lower-SES students can realize comparable gains.

Source:

<https://documents1.worldbank.org/curated/en/099548105192529324/pdf/IDU-c09f40d8-9ff8-42dc-b315-591157499be7.pdf/>

¹⁰⁰ <https://documents1.worldbank.org/curated/en/099548105192529324/pdf/IDU-c09f40d8-9ff8-42dc-b315-591157499be7.pdf>

Tertiary & Vocational Education

Current Situation / Gaps: Several recent investigations signal that Nigerian higher education institution, universities, polytechnics and colleges, are still at early stages of integrating artificial intelligence (AI) into teaching, learning and institutional practice. One study found only “moderate awareness” of AI among students and faculty, with very few dedicated AI-programmes or broad curriculum adoption.¹⁰¹ Another review surveying 74 published articles on Nigerian AI-in-education between 2008 and 2022 notes that while AI applications such as student-performance analytics and e-learning platforms are emerging, more advanced systems like intelligent tutoring, machine-learning modules and deeply interdisciplinary AI programmes remain rare.¹⁰² These findings illustrate a readiness gap: policy momentum and research interest exist, but the substantive translation into tertiary curricula, teacher development and institutional strategy is still limited.

In the technical and vocational education and training (TVET) domain, the pattern is similar but layered with additional challenges. For example, studies highlight that TVET institutions often rely on traditional curricula, face low instructor capacity for AI tools, and lack unified institutional strategies to introduce context-appropriate AI content.¹⁰³

Infrastructure constraints are particularly acute in these settings: many TVET facilities operate with limited connectivity, inadequate devices, and minimal support for hands-on, AI-driven experiential learning. Without these enablers, the shift from theoretical AI awareness to applied, vocationally relevant AI skills remains weak.

In terms of policy recognition, the National TVET Policy and Strategy on Open, Distance, Flexible and e-Learning (ODFeL) (2018) for Nigeria provides a strong foundation for ICT-enabled learning but is neither AI-integrated nor AI-responsive. The policy focuses on digital literacy, e-learning platforms, ICT infrastructure, virtual learning environments, and flexible delivery for access and equity; however, it contains no references to artificial intelligence, machine learning, adaptive learning, or the use of AI-powered tools in instruction, assessment, teacher training, curriculum development, or quality assurance. Technology priorities remain limited to basic digital systems, computers, software, connectivity, and OER (OER) without recognition of AI as a transformative component of future skills development or workforce competitiveness. While the policy acknowledges rapid technological change and the need for ongoing staff capacity building, these commitments do not translate into AI-specific competencies, pathways, or governance frameworks, leaving a critical gap in preparing Nigeria’s TVET sector for AI-driven labour markets.

¹⁰⁰ <https://documents1.worldbank.org/curated/en/099548105192529324/pdf/IDU-c09f40d8-9ff8-42dc-b315-591157499be7.pdf>

¹⁰¹ <https://iiardjournals.org/get/WJIMT/VOL.%209%20NO.%205%202025/Artificial%20Intelligence%20Awareness%20101-114.pdf>

¹⁰² <https://link.springer.com/article/10.1007/s44163-024-00163-y>

¹⁰³ <https://earnia.org/e4e356ef6f4f9e939e06f04e06f60ef3ef0560f0de6/AN%20ANALYSIS%20OF%20THE%20ROLE%20OF%20ARTIFICIAL%20INTELLIGENCE%20IN%20NIGERIA.pdf>

TABLE 3 National TVET Policy and Strategy on ODFeL is AI-integrated or AI-sensitive:

Policy Aspect	Status in TVET ODFEL Policy (2018)	Comments
Artificial Intelligence (AI) Explicitly Present	Not Mentioned	No reference to AI, machine learning, or AI-powered systems in policy or objectives
Digital Literacy / ICT Integration	Included	Strong emphasis on ICT, e-learning, digital skills as foundations
Adaptive/Personalized Learning (AI-driven)	Not Present	Does not include AI -based adaptive learning platforms or strategies
Teacher Training in AI	Not Present	Focus on ICT upskilling for teachers, not AI literacy or pedagogy
EdTech & Technology in Content Delivery	Included (General)	E-learning, OER, virtual labs planned, but not AI -enhanced or intelligent systems
AI Ethics, Bias, and Data Privacy	Absent	No provisions for AI ethics, bias, responsible or ethical AI use, or AI data privacy
Facilitation of AI Upskilling/Workforce Pathway	Absent	No pathways for AI upskilling or workforce development outlined
Policy/Strategy Future Readiness (AI)	Partial (ICT-ready)	ICT infrastructure foundation could support AI in future, but currently not AI-focused

Source: The National Board for Technical Education

Equity, governance and institutional capacity deficiencies further complicate the picture. For example, one study investigating generative AI adoption across Nigerian universities found socio-cultural and technical factors, such as tool usability, community norms, and support systems, strongly influenced student engagement, with students in less well-resourced institutions facing greater barriers.¹⁰⁵ Moreover, the absence of coherent institutional frameworks for AI-teaching, assessment practices revised for AI-literacy, data-privacy protocols and industry linkages means that even where AI modules exist, their impact may be uneven. In short, while Nigerian tertiary and vocational systems show promising early steps toward AI integration, substantial systemic work remains to establish scalable, equitable and high-quality AI-capable programmes.

Global Case Study / Best Practice: A study on AI's impact in higher education reported that AI interventions improved personalised learning, assessment efficiency and skill-development for future work.¹⁰⁶ For vocational pathways, programmes that integrate AI with domain-specific trades (e.g., agriculture, manufacturing, health technology) demonstrate how students can gain applied AI skills aligned to the labour market. Nigeria's framework could leverage these models by developing tertiary/vocational AI modules (technical AI, domain-specific AI), and include inclusive entry mechanisms for marginalised learners.

¹⁰⁴ <https://www.nbte.gov.ng/nbte/sites/default/files/2024-01/National%20TVET%20Policy%20and%20Strategy%20on%20ODFeL.pdf>

¹⁰⁵ <https://slejournal.springeropen.com/articles/10.1186/s40561-024-00352-3>

¹⁰⁶ <https://files.eric.ed.gov/fulltext/EJ1384682.pdf>

1. **Ivy Tech Community College, USA:** The college deployed an AI-driven predictive analytics tool (dubbed NewT) using Google Cloud and machine-learning to analyse student engagement data (e.g., LMS log-ins, attendance, assignment submission) and flag at-risk students within the first two weeks of a semester.¹⁰⁷ During the pilot, they processed approximately 10,000 course sections and identified about 16,000 at-risk students; targeted interventions helped roughly 3,000 of them avoid failing and raise their grades to a C or better.¹⁰⁸ Key learnings: early detection plus holistic support (academic + non-academic) is essential; data-driven insights must be coupled with responsive human action; and robust privacy-governance is necessary.
2. **University of Melbourne, Australia:** Research at the University of Melbourne highlights how students and faculty approach generative AI and predictive analytics in higher education. A 2023 study found that many students are

cautious about adopting AI tools, many academics believe their institutions are not yet ready for generative AI integration.¹⁰⁹ The key insight for policy: AI systems must not only be technically ready but also embedded in transparent institutional frameworks, aligned assessments, and professional development for faculty, otherwise uptake is slow and contested.

3. **UNITAR International University / Malaysian TVET Context:** A qualitative study of AI in Malaysian TVET and STEM programmes found that AI tools (adaptive learning, intelligent tutoring) hold promise for enhancing learning efficiency, critical thinking and future-work readiness.¹¹⁰ However, the study also uncovered major barriers: instructor readiness, infrastructure deficits, and weak alignment of AI modules with vocational and industry needs. This suggests that vocational programmes must combine AI content with authentic, domain-specific applications and strong industry linkages

Box 4: Implications for Nigeria

From these cases we draw the following insights relevant to Nigeria's tertiary and TVET sectors:

- Early warning / predictive analytics can significantly boost student success in higher education if deployed with strong human-support systems.
- Institutional readiness (policy, faculty training, governance) is just as important as technology itself.
- Vocational programmes must tie AI content to real-world industry contexts and ensure instructors, infrastructure and local relevance are in place.

¹⁰⁷ https://edu.google.com/intl/ALL_us/resources/customer-stories/ivytech-gcp/

¹⁰⁸ <https://www.globaladmissions.com/blog/how-universities-used-ai-to-revolutionize-higher-education-experiences>

¹⁰⁹ <https://pursuit.unimelb.edu.au/articles/university-students-are-using-ai-but-not-how-you-think>

¹¹⁰ <https://spm-online.com/jrmste/index.php/journal/article/view/15>

Non-Formal & Adult Education

Current Situation / Gaps: In Nigeria, adult and non-formal education pathways are significantly under-prepared for the AI-era. Studies show that most programmes do not offer structured opportunities to learn about artificial intelligence, machine learning or even foundational digital literacy. For example, research indicates that “adult and non-formal education programmes are intricately interwoven ... and while ICT tools are emerging, the realization of their potential is a series of pedagogical and practical challenges.”¹¹¹ Without even basic digital competencies, adult learners face a first-mile barrier that prevents further progress into AI-relevant learning pathways.

The digital literacy deficit among adult learners is wide and well-documented: more than half of Nigeria’s population lack digital skills.¹¹² Furthermore, infrastructure constraints are acute, only about 9.4 % of households own a computer and 34.6 % have home internet access.¹¹³ These structural limitations—lack of devices, unreliable electricity, unaffordable connectivity, undermine the ability of adult learners to engage with digital tools, let alone adaptive or AI-powered platforms.

The teaching and institutional architecture in adult and non-formal settings is also weak. Facilitators often lack training in digital pedagogy and the programming of adult education to support new learning forms. One study reports that adult education facilitators must become digitally oriented to handle new technologies in effective teaching, yet many remain unprepared.¹¹⁴ This gap means that instructional design and delivery rarely support adult learners to move from basic literacy

to complex AI-skills. Moreover, policy frameworks remain generic: while there is a national digital literacy framework, it does not explicitly mandate AI literacy, personalised learning or modular up-skilling for adults.¹¹⁵

Flexibility and relevance are insufficient in many adult/non-formal programmes. Learning modules often fail to accommodate working adults, caregivers or marginalized learners requiring self-paced or modular formats. Many courses are not context-tailored (e.g., not in local languages or not aligned with local labour-market needs), and certification mechanisms are weak. Research shows that distance and e-learning models exist but are under-scaled, and adult education remains functionally disconnected from evolving digital-skills and AI-upskilling demands.¹¹⁶ In sum, while adult and non-formal education is crucial for inclusive human-capital development in Nigeria, the sector currently lacks foundational, structural and policy readiness to support AI-driven learning trajectories.

A review of the 2017 Policy Guidelines for Mass Literacy, Adult and Non-Formal Education (NMEC found that note AI-integrated. Nowhere do they reference artificial intelligence, machine learning, intelligent tutoring, adaptive assessment, or algorithms. Their technology lens is limited to general ICT radio/TV, computers, internet, mobile-phone literacy, and basic digital skills without curriculum pathways for AI literacy or provisions for facilitator training on AI use, data ethics, or personalized/adaptive learning. Research and innovation sections likewise omit AI, signaling a gap between current adult-learning goals and the competencies demanded by an AI-enabled economy.

¹¹¹ <https://pdfs.semanticscholar.org/30ae/0d0e30ab7ee727b4920f6aefdc390b2e71a9.pdf>

¹¹² <https://businessday.ng/technology/article/over-50-of-nigerias-population-lack-digital-skills-world-bank/>

¹¹³ <https://thedocs.worldbank.org/en/doc/a607bb6e3b76d2be0f3db8db34dcf73e-0140022025/original/1Nigeria-TFOC2441-Digital-Skills-Report-final.pdf>

¹¹⁴ <https://rsisinternational.org/journals/ijriss/articles/digital-literacy-and-its-implications-for-sustainable-adult-education-in-the-21st-century/>

¹¹⁵ <https://nitda.gov.ng/wp-content/uploads/2023/07/Digital-Literacy-Framework.pdf>

¹¹⁶ <https://ibadanadulthoodeducationjournals.com.ng/media/2022/08/Best-Practices-of-Promoting-Adult-and-Non-Formal-Education-via-Distance-and-E-Learning-in-Nigeria.pdf>

TABLE 4 NMEC AI Integrated Status

Aspect	Status in Policy	Comments
Artificial Intelligence	Not Integrated	No mention, no curriculum, no strategy
Digital/ICT Literacy	Included	Radio, TV, computers, internet, some mobile use
Adaptive Learning/EdTech	Potential	Only via generic ICT, e-learning, VSAT, not AI-driven
Facilitator AI Training	Absent	Digital upskilling, yes; AI/ML, no
Data/Research	Supports ICT, not AI	Management Information Systems, but no AI/data pipeline
Innovation/Future Readiness	Partial	Policy could be revised to add AI pathways

Source: NMEC 2017¹¹⁷

That said, the policy is partially AI-friendly in the sense that it embraces digital literacy, e-learning, mobile applications, ICT centres, VSAT-enabled virtual classrooms, and education-management information systems, elements that could underpin future AI adoption. To move from ICT-readiness to genuine AI-readiness, an update should explicitly add: (1) AI literacy outcomes and modular upskilling pathways for adults; (2) facilitator PD on AI pedagogy, data protection, and responsible use; (3) standards for AI-powered instruction and assessment (including accessibility and inclusion); and (4) governance for ethical, data-driven, adaptive systems. In short: the policy provides a digital foundation, but it requires substantive revision to support AI-enabled, inclusive lifelong learning.

Global Case Study / Best Practice: Global experience shows that when AI tools are designed around flexibility, personalization, and inclusion, they can significantly improve adult learning outcomes. China has successfully incorporated intelligent tutoring systems into adult education since 2015, providing adaptive literacy, numeracy, and digital-skills learning that adjusts to pace and need while giving facilitators real-time analytics to monitor progress. Evaluations report higher retention and achievement among working adults who require self-paced learning options.

Similarly, UNESCO promotes AI-enhanced lifelong learning through models such as voice chatbots, AI-supported mobile learning, and OER platforms with algorithmic recommendations, especially

targeting marginalized learners, including those in low-connectivity areas. In Bangladesh, mobile-first programmes for women use AI-powered quizzes and vernacular content to strengthen confidence and accelerate literacy gains. Global virtual-learning providers such as MyOneFlow and ModernCampus also use adaptive assessments and accessibility tools that have improved outcomes for non-traditional learners, including those with disabilities.

Consumer-scale platforms demonstrate another layer of innovation. Tools like Khanmigo and Duolingo Max show how AI can provide guided support, conversational practice, and instant comprehension feedback, allowing adults to learn in small bursts that fit around work and caregiving demands.¹¹⁸ Meanwhile, international upskilling initiatives, such as UNESCO's Global Skills Academy is targeting 10 million youth and adults by 2029, already reaching 850,000+ across 60+ countries, offering modular digital/AI skills via partners.¹¹⁹ IBM SkillsBuild provides free AI courses with digital credentials aimed at jobseekers and career-switchers, including Africa-focused offerings and a public commitment to train 2 million people in AI over three years. Microsoft + LinkedIn run widely used Generative AI learning paths (prompting, Copilot use, productivity), giving adult learners short, stackable routes into AI-enabled work.¹²⁰ These models show how public-private coalitions can meet adults where they are online, credentialed, and employment-linked.¹²¹

¹¹⁷ <https://education.gov.ng/wp-content/uploads/2019/12/POLICY-GUIDELINES-FOR-MASS-LITERACY-ADULT-NON-FORMAL-EDUCATION.pdf>

¹¹⁸ <https://www.khanmigo.ai/>

¹¹⁹ <https://www.unesco.org/en/articles/unescos-global-skills-academy-spotlight-session-worldskills-conference-2024>

¹²⁰ <https://skillsbuild.org/adult-learners>

¹²¹ <https://skillsbuild.org/adult-learners/explore-learning/artificial-intelligence>

Across these examples, a clear set of transferable principles emerges:

1. adaptive micro-learning that supports flexible schedules;
2. a mobile-first approach that reduces infrastructure barriers;
3. culturally and linguistically relevant content;
4. AI analytics that enable facilitators to target support; and
5. robust data-privacy and ethical safeguards to ensure inclusion.

These practices show how non-formal systems can progress from basic ICT exposure toward AI-ready and employment-linked lifelong learning, particularly for women, rural populations, and other marginalized groups.

5.5

Girls Education & Inclusion Across All Levels

Current Situation / Gaps: Across Nigeria, the gendered access gap to the digital world remains the steep hurdle that any AI-in-education agenda must clear. Recent analyses show a persistent disparity in women's internet use (roughly 20% of women vs 37% of men in one national brief) and continuing smartphone and mobile-internet gaps, which constrain girls' and women's ability to engage with AI-enabled learning inside and outside school.¹²² Newer sector snapshots indicate only 39% of Nigerian women own a smartphone compared with 50% of men, underscoring affordability and access barriers that would blunt the reach of AI-supported learning, especially for lower-income and rural learners¹²³

Participation bottlenecks are equally visible in STEM pathways that feed into AI. Evidence from secondary and tertiary assessments shows female

participation in core subjects such as Physics often falls below 30% in many states, with long-standing under-representation continuing into higher education and technical programmes.¹²⁴ This supply-side constraint interacts with digital-safety risks: technology-facilitated gender-based violence (TF-GBV), from image-based abuse to cyberstalking, discourages girls and women from online spaces that are increasingly central to AI-supported learning and collaboration.¹²⁵ Without safeguards and confidence-building measures, AI rollouts risk reproducing offline inequities in digital form.¹²⁶

There are promising, but partial, institutional responses. UNESCO notes the growing use of state-level "gender desks" to make schools safer and keep marginalised students learning, yet these mechanisms are not yet systematically linked to AI-specific access, safety, or participation strategies (devices, data subsidies, protected digital spaces, or girls' AI clubs and mentors).¹²⁷ Broader affordability pressures highlighted by global surveys of women entrepreneurs who cite data cost and online harassment compound the problem and can mute the benefits of AI-enabled learning for girls across basic, secondary, tertiary, and non-formal settings. To avoid widening the gap as AI tools enter classrooms, inclusion measures must be designed in from the start and tracked with gender-disaggregated data.¹²⁸

Global Case Study / Best Practice: Global programmes that centre girls' agency, real projects and recognised credentials show strong results. Girls Who Code reports reaching about 760,000 learners, including 330,000 college- or career-aged alumni; alumnae earn CS-related degrees at roughly five times the U.S. average and the organisation is targeting gender parity in entry-level CS roles by 2030.¹²⁹ Technovation Girls likewise documents long-run STEM engagement and has expanded AI-focused curricula and family programmes that position girls as creators building apps and AI prototypes rather than passive users.¹³⁰ These models combine hands-on projects, mentorship and micro-credentials,

¹²² https://banyanglobal.com/wp-content/uploads/2023/08/Nigeria-GDD-Brief_Final-508-May-2023.pdf

¹²³ https://banyanglobal.com/wp-content/uploads/2023/08/Nigeria-GDD-Brief_Final-508-May-2023.pdf

¹²⁴ <https://iiardjournals.org/get/IJASMT/VOL.%2011%20NO.%206%202025/STEM%20Education%20and%20Gender%2043-55.pdf>

¹²⁵ <https://www.unfpa.org/TFGBV>

¹²⁶ <https://iiardjournals.org/get/IJASMT/VOL.%2011%20NO.%206%202025/STEM%20Education%20and%20Gender%2043-55.pdf>

¹²⁷ <https://www.iiep.unesco.org/en/articles/gender-desks-frontline-girls-education-nigeria>

¹²⁸ <https://www.theguardian.com/global-development/2025/mar/20/cost-data-developing-world-digital-women-in-business-report-female-entrepreneurs-internet-access>

¹²⁹ <https://girlswhocode.com/news/2024-annual-report>

¹³⁰ <https://girlswhocode.com/2024report/>

illustrating a transferable playbook for Nigeria's JSS–SSS pathways and after-school clubs.¹³¹

At system level, multilateral initiatives provide scaffolding for scale and equity. UNESCO highlights that women constitute only about 35% of global STEM graduates and remain under-represented in AI roles, underscoring the need for early, sustained, gender-responsive pathways.¹³² ITU's latest figures show a persistent digital gender gap around 70% of men versus 65% of women online globally, with wider gaps in least-developed contexts so inclusive AI learning must bundle devices, affordable data and safe online spaces with instruction.¹³³ In parallel, workforce-linked programmes (e.g., UNESCO's Global Skills Academy, IBM SkillsBuild, Microsoft/LinkedIn GenAI paths) provide modular, stackable credentials that help adult and adolescent girls transition into AI-enabled jobs.^{134,135}

Common best-practice themes emerge across contexts: targeted digital/AI training for girls paired with mentorship and safe learning environments; teacher professional development that addresses bias and enables inclusive pedagogy; public–private partnerships that supply devices, content and industry exposure; and rights-based, community-anchored delivery that integrates socio-emotional support and GBV safeguards. Adaptive, mobile-first technologies (including offline-capable tablets and AI tutors) enable individual pacing and local-language content, while gender-disaggregated monitoring ensures gaps close rather than widen.

Nigeria's recent efforts echo these lessons. The World Bank-supported "Gina Mata, Gina al-Umma" programme trained 1,295 disadvantaged girls

across six northern states, blending digital skills, gig-economy readiness and rights-based empowerment.^{136,137} Edo State's "AI for Girls" initiative with Elite Global AI began with 1,000 secondary-school girls and plans expansion across universities and states.¹³⁸ MindAfrica's work in Rivers State has engaged girls with AI, coding, robotics and mobile-app workshops alongside mentorship and online-safety awareness.¹³⁹ In the Microsoft Copilot adaptive-tutoring pilot (>750 students), girls improved at roughly twice the rate of boys, narrowing achievement gaps and boosting confidence and participation.¹⁴⁰

Inclusive design and targeted investment are essential to ensure that AI integration advances, rather than reinforces, gender inequality. Popoola's success in applying AI suggestions while relying on her grandfather's indigenous knowledge underscores how AI can empower girls and deepen locally contextual innovation.¹⁴¹ This real-world illustration shows how culturally grounded examples can motivate girls to view AI not as distant technology but as a supportive, relatable learning tool.

¹³¹ https://girlswhocode.com/assets/downloads/craft-prod/downloads/GWC_Impact_Evaluation_Study_Final_Report.pdf

¹³² <https://www.educationtoday.co/news/daily-news/unesco-flags-gender-gap-in-stem-women-only-35-of-graduates-just-26-in-ai-workforce>

¹³³ <https://www.educationtoday.co/news/daily-news/unesco-flags-gender-gap-in-stem-women-only-35-of-graduates-just-26-in-ai-workforce>

¹³⁴ <https://www.itu.int/itu-d/reports/statistics/2024/11/10/ff24-the-gender-digital-divide/>

¹³⁵ <https://www.globalgiving.org/projects/digital-skills-for-200-nigerian-girls/>

¹³⁶ <https://thedocs.worldbank.org/en/doc/c77a3a372cd9f923d9fc39e187820dfa-0140022025/original/LEAS-2-Presentations.pdf>

¹³⁷ <https://blogs.worldbank.org/en/nasikiliza/transforming-futures-digital-skills-for-girls-and-women-in-northern-nigeria>

¹³⁸ https://www.linkedin.com/posts/vwakpor-efuetanu-2b7274240_we-just-signed-a-partnership-that-will-shape-activity-7373266859598622720-C-xc

¹³⁹ <https://www.globalgiving.org/projects/digital-skills-for-200-nigerian-girls/>

¹⁴⁰ <https://documents1.worldbank.org/curated/en/099548105192529324/pdf/IDU-c09f40d8-9ff8-42dc-b315-591157499be7.pdf>

¹⁴¹ <https://drive.google.com/file/d/1qGjrtkp1Z2n25A2i8XNBPZUqW9L7xW1/view>

6.

Teacher Capacity Building

Current Situation Analysis / Gaps: In Nigeria, teachers' readiness to integrate AI and advanced digital technologies into their practice remains uneven and constrained by systemic barriers. A 2025 study of Nigerian teachers' readiness to adopt AI in education found that, while awareness is growing, access to infrastructure, digital resources and institutional support remain major limitations, particularly in rural and under-resourced settings.¹⁴² In one Yobe State study of biology teachers, although willingness to integrate AI was high, the positive outcomes were heavily dependent on the availability of computers and internet access, indicating that enthusiasm alone is insufficient without operational capacity.¹⁴³ Surveys such as Ayanwale et al. (Nature 2025) reveal that while teachers are motivated and recognise the potential of AI for learning, they also feel under-prepared and unsupported. Without a coordinated national strategy that links device access, teacher certification, rural inclusion, ethical standards and mentorship, adoption of AI pedagogy is likely to remain slow, uneven and inequitable.

"To our committed teachers, you are at the heart of this transformation. The knowledge and skills you gain from this training will create a ripple effect, equipping students for a future where digital and AI competencies are essential. I urge you to seize this opportunity wholeheartedly and apply what you learn to enrich classroom teaching and learning."
Prof. Suwaiba Ahmed,
represented by Dr. Clara Ujam,
2025

Most teachers have not received formal training in AI concepts, the integration of AI into classroom instruction, or the use of AI-powered instructional tools. Existing teacher education programs remain focused on basic ICT competence, and rarely include modules on AI literacy, ethical AI use, or project-based digital pedagogies. Structurally, only about 12% of federal universities currently offer a dedicated AI course aimed at teacher or pre-service educator cohorts, and public education funding to support digital and AI teacher development remains under 10% of the national budget. Many schools still lack functional ICT labs or modern devices, making meaningful AI integration impractical.

Professional development is expanding but is not yet aligned with the demands of AI-enhanced instruction. Studies show that many in-service and pre-service teacher programmes in Nigeria focus on general ICT or computer-use competencies, rather than on AI-specific pedagogical design, prompt engineering, data-driven feedback, or ethical considerations of AI deployment.¹⁴⁴ Furthermore, teachers report needing sustained, scaffolded training and ongoing mentoring to translate new skills into classroom practice a gap confirmed by research on teacher educators who noted infrastructure, training and ethics as key barriers to AI adoption.¹⁴⁵

Another critical challenge lies in the institutional and instructional architecture: many schools lack leadership support, forums for collaborative planning, and monitoring systems to guide teacher integration of AI tools. Rural schools in particular face outdated equipment and unreliable connectivity, conditions that make it difficult to deploy AI tools, which often require higher data and computing capacity. Research indicates that teacher readiness is closely linked to these infrastructural and organisational supports.¹⁴⁶

¹⁴² <https://www.nature.com/articles/s41598-025-08018-x>

¹⁴³ <https://www.seahipublications.org/wp-content/uploads/2025/04/IJISSEER-J-6-2025.pdf>

¹⁴⁴ <https://journals.openedition.org/ries/17323>

¹⁴⁵ <https://dergipark.org.tr/en/download/article-file/4014590>

¹⁴⁶ <https://iiardjournals.org/get/IJCSMT/VOL.%2011%20NO.%206%202025/The%20Impact%20of%20AI%20on%20STEM%2074-86.pdf>

Box 5: Linking Classroom Practice to AI Pedagogy

In *The Master not the Machine – Lessons in AI Literacy* (Olasupo & Adeoti, 2025), Mr. Adebayo's classroom offers a practical example of how teachers can guide students to use AI responsibly. He teaches learners to verify AI outputs, question digital information, and rely on human judgment before accepting algorithmic answers.

This mirrors the Framework's call for teacher capacity building that embeds ethical reflection, human-in-the-loop practice, and learner-centred facilitation. Through such examples, teachers are positioned not as passive users of technology but as critical mediators who help students become discerning, creative, and ethical digital citizens.¹⁴⁷

In terms of teacher preparation, a 2025 study by SEAH Publications on AI in Nigerian teacher education found that the lack of training (mean score = 3.62) and poor infrastructure (mean = 3.55) were identified as the most critical barriers followed by limited funding and weak policy support.¹⁴⁸ In response, the Federal Government's Zero-Rated Data and Mobile Devices Initiative, launched in October 2025, aims to reach 8,000 teachers across all six geopolitical zones, granting free data access to digital learning tools and AI-enabled teaching platforms.¹⁴⁹ This program is aligned with broader national goals to integrate AI tutors, curated digital content and blended-learning analytics into the national eLearn platform. Early signals of student engagement and uptake are positive: Google search trends in 2025 show a 60% increase in education-related AI queries and a 200% surge in "AI + studying" search terms.¹⁵⁰ Comparative pilot studies in states like Edo show AI-assisted instruction boosting student test performance by 15-22%.¹⁵¹ These findings underscore that while demand and potential are growing, delivery infrastructure, teacher capacity and governance frameworks must be strengthened to translate AI readiness into meaningful, equitable adoption.

Best Practice / Case Studies

There are emerging models of teacher capacity building for AI in education that show promise.

Singapore – Systemwide PD with AI focus. Singapore couples national policy with sustained professional development so teachers can actually use AI in planning, assessment, and differentiation. In the latest TALIS 2024 release, 76% of Singaporean teachers reported participating in PD on using AI for teaching and learning (OECD avg: 38%), reflecting deliberate, centrally supported upskilling and school-level enablement.¹⁵²

United Kingdom – Oak National Academy's AI lesson assistant (Aila). England's curriculum body Oak is trialling "Aila," an AI planning assistant that generates standards-aligned lesson sequences via RAG over a vetted content bank, with governance published in the UK's algorithmic transparency record. Early insights suggest reductions in planning time and opportunities to strengthen pedagogical intent when teachers remain "in the loop."¹⁵³

¹⁴⁷ <https://drive.google.com/file/d/1qGjrtkp1Z2n25A2i8XNBCpZUqw9L7xW1/view>

¹⁴⁸ <https://www.seahipublications.org/wp-content/uploads/2025/10/IJIISTR-D-5-2025.pdf>

¹⁴⁹ <https://www.thisdaylive.com/2025/10/23/fg-launches-scheme-to-empower-teachers-with-mobile-devices-free-data/>

¹⁵⁰ <https://guardian.ng/technology/more-nigerian-students-embracing-ai-says-google-data/>

¹⁵¹ <https://blogs.worldbank.org/en/developmenttalk/addressing-the-learning-crisis-with-generative-ai--lessons-from->

¹⁵² <https://www.moe.gov.sg/news/press-releases/20251007-singapore-teachers-embrace-digital-technologies-and-benefit-from-strong-professional-development-oecd-talis-2024-study>

¹⁵³ <https://www.gov.uk/algorithmic-transparency-records/oak-naddditional-academy-aila-oaks-ai-lesson-assistant>

UNESCO – AI Competency Framework for Teachers (AI-CFT). To move beyond generic ICT training, UNESCO’s new AI-CFT defines 15 competencies across five dimensions (human-centred mindset, ethics, AI foundations, AI pedagogy, and AI for professional learning). Countries and providers use it to align PD content, certification, and classroom practice, closing the gap between “tool familiarity” and pedagogical use of AI.¹⁵⁴

Finland demonstrates a contrasting, bottom-up model that prioritises teacher voice and ethical reflection. AI literacy and responsibility are embedded in teacher PD programmes, and educators are encouraged to co-design and trial AI tools before adoption. Collaboration between teachers’ unions, universities, and policymakers ensures that AI strengthens, rather than replaces, professional judgement, supporting creativity, inclusion, and learner agency.

China has taken a coordinated, system-wide approach: AI teaching modules are mandatory for educators across grades, complemented by government-funded resource hubs, AI textbooks, and ongoing professional development.¹⁵⁵ This top-down strategy has rapidly expanded teacher competence in AI concepts and classroom use. However, observers stress the need to address concerns around data governance, teacher autonomy, and surveillance as integration deepens.

Nigeria (Edo State) – Teacher-facilitated GenAI tutoring pilot. In a World Bank RCT, after-school AI tutoring (GPT-4 via Copilot) delivered 0.31 σ composite gains in six weeks, with teachers acting as facilitators and guardrails (curriculum alignment, responsible use). The study demonstrates that targeted teacher roles plus lightweight training can unlock sizeable learning effects, even with bandwidth and power constraints.¹⁵⁶

Nigeria (Federal) – AI PD at national scale beginnings. In 2025, the Federal Government launched AI training for 6,000 senior-secondary teachers across all states via the National Senior Secondary Education Commission, an early but concrete step toward mainstream teacher upskilling that can be expanded and tied to certification.¹⁵⁷

Cautionary tale – South Korea’s AI textbooks. South Korea’s rapid rollout of AI-driven digital textbooks faced teacher/parent backlash and was rolled back in 2025, with critiques citing inadequate preparation, workload concerns, and privacy risks. The episode underscores a core lesson: invest first in teacher PD, choice, governance, and staged evaluation, then scale.¹⁵⁸

Blend Singapore’s sustained PD model and UNESCO’s AI-CFT with Oak’s teacher-in-the-loop design; pair these with Nigeria’s emerging national PD (the 6,000-teacher cohort) and the Edo pilot’s facilitation model. All these provides a blueprint: set clear competencies, fund ongoing PD, publish governance, keep teachers central, and scale only after classroom-level evidence and support systems are in place.

¹⁵⁴ <https://www.unesco.org/en/articles/ai-competency-framework-teachers>

¹⁵⁵ <https://www.forbes.com/sites/cathyrubin/2025/08/26/ai-in-education-best-practices---whats-working-in-classrooms-around-the-globe/>

¹⁵⁶ <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099548105192529324>

¹⁵⁷ <https://punchng.com/fg-unveils-ai-training-for-6000-teachers-nationwide/>

¹⁵⁸ <https://www.forbes.com/sites/cathyrubin/2025/08/26/ai-in-education-best-practices---whats-working-in-classrooms-around-the-globe/>

7.

Infrastructure and Equitable Access

Current Situation Analysis / Gaps: Nigeria's AI-in-education ambitions run headlong into basic access constraints. Connectivity is the most binding: recent NCC figures show only 23% of rural communities are online (vs 57% in urban areas), underscoring a widening urban–rural divide that directly limits any data-dependent AI tools in schools. For example, one study identified that the digital divide remains “a significant barrier to educational equity in Nigeria ... hindering the potential for technology-driven learning and socio-economic advancement.” Power supply compounds the gap, Freedom House notes 91.4% of urban residents have electricity versus 30.4% in rural areas, meaning even connected schools struggle to keep devices and routers powered. These structural factors make routine use of cloud-based AI (tutors, analytics, content generation) unreliable outside major cities.¹⁶⁰

School-level infrastructure is uneven and, in many places, minimal. A 2025 snapshot reported that only 17% of schools have reliable internet, with many rural schools lacking ICT labs entirely, limiting both teacher PD and classroom use of AI. Federal moves to “digitalise public schools” with smart boards and smart-learning

initiatives are promising but nascent; they will require back-end bandwidth, device maintenance, and local support to translate hardware drops into sustained AI-supported pedagogy. The Universal Service Provision Fund (USPF) continues to finance connectivity and access programmes for underserved areas, yet coverage at school level remains far from universal.¹⁶¹ Disparity between urban and rural schools is stark: many rural schools lack basic ICT infrastructure, stable power and internet access, making any AI-enabled learning difficult. A blog post noted:

“In many public schools nationwide, particularly those in rural areas, have little to no access to computers, depriving students of the opportunity to develop essential digital skills.”
Prof. Suwaiba Ahmed,
represented by Dr. Clara Ujam,
2025¹⁶²

The story of Popoola using limited ICT resources in Ilorin to diagnose a crop disease illustrates both the potential and the infrastructure challenges of AI use in rural schools. Her experience shows that even when technology is available, limited devices and intermittent access can restrict students' opportunities to fully benefit from AI-enabled learning.¹⁶³

Affordability and inclusion are persistent cross-cutting issues. Data costs remain a major barrier for households and teachers, and the gender digital gap is sizable: women's internet use and smartphone access trail men's, with multiple

¹⁶⁰ <https://www.vanguardngr.com/2025/10/only-23-of-rural-communities-have-internet-access-in-nigeria-ncc/>

¹⁶¹ <https://thenigeriaeducationnews.com/2025/07/01/the-silent-breakdown-in-nigerias-classrooms-no-labs-no-libraries-no-digital-tools/>

¹⁶² <https://businessday.ng/interview/article/how-poor-infrastructure-skill-gap-hinder-ai-education-technology-education-ajala/>

¹⁶³ <https://drive.google.com/file/d/1qGjrtkp1Z2n25A2i8XNBCpZUqw9L7xW1/view>

studies documenting lower female usage rates and affordability constraints, gaps that risk being mirrored in AI access and skill acquisition for girls. Without targeted subsidies, offline/low-bandwidth options, and safety measures, AI rollouts could deepen inequities by gender, geography, and income.¹⁶⁴ Although there are CSR and private-sector initiatives, these tend to be fragmented, often concentrated in urban or better-resourced schools. The systemic scaling of affordable AI devices and connectivity through partnerships remains weak.

Implications for the AI-Education stack. These constraints affect every layer of the AI learning ecosystem:

- i. cloud-first AI tutors and analytics dashboards need stable power/bandwidth;
- ii. device scarcity limits 1:1 or even 1:3 usage models;
- iii. teacher PD can't rely on always-online platforms; and
- iv. equity goals require mobile-first, low-data, offline-capable AI tools, plus public access points (community ICT hubs) and school-based caching.

Short-term programmes, like zero-rated access and device pilots—help, but the system will need sustained investment in rural backhaul, school Wi-Fi, power backup, device lifecycle/repair, and gender-responsive affordability policies to make AI in education both useful and fair at scale.¹⁶⁵ The bottom line is that, Nigeria has clear momentum (smart-classroom initiatives; USPF connectivity/access programmes), but the binding constraints, rural connectivity at 23%, unreliable electricity in rural areas, low school-level internet readiness, and affordability/gender gaps must be tackled head-on or AI in education will remain a patchwork of pilots rather than a nationwide, equitable capability.

Infrastructure sustainability is a major issue: schools may receive devices or connectivity once, but ongoing maintenance, upgrades, teacher training, and refurbishment are rarely budgeted. Studies highlight that without accompanying support and sustainability planning, initial digital interventions lose impact. For example, a paper on early childhood digital inclusion in Nigeria noted the need for “contextualised, culturally appropriate, and long-lasting approaches ... ensuring that young learners in the foundational level ... do not miss out” because of infrastructure and access constraints.¹⁶⁶



¹⁶⁴ https://banyanglobal.com/wp-content/uploads/2023/08/Nigeria-GDD-Brief_Final-508-May-2023.pdf

¹⁶⁵ <https://ncc.gov.ng/about/servicom-charter/departamental-charters/universal-service-provision-secretariat-service-charter>

¹⁶⁶ <https://www.seahipublications.org/wp-content/uploads/2025/07/IJIISTR-S-15-2025.pdf>

Case Study / Best-Practice:

- **Public access funds as the backbone for last-mile connectivity.** Nigeria's Universal Service Provision Fund (USPF) was created to finance network rollout in unserved and underserved communities, including schools and tertiary campuses (e.g., the Campus Connect programme). The USPF model shows how a ring-fenced levy can underwrite connectivity, labs and shared services that AI-enabled teaching depends on power, backhaul, and device pools, while targeting equity from the outset. For AI adoption, the lesson is to use USPF-style mechanisms to prioritize school clusters and teacher-training hubs in connectivity deserts, pairing grants with service-level and maintenance obligations.¹⁶⁷
 - **Pilot-to-scale infrastructure lessons from the Edo State GenAI trial.** The World Bank RCT in Edo State proved that virtual tutoring can deliver large learning gains despite real-world frictions (power cuts, patchy internet). Operational notes from the pilot emphasize practical mitigations scheduled sessions, supervised labs, cached prompts/materials, and simple devices, pointing to an access playbook for lower-resource schools: stabilize power/Internet where feasible, and design the learning flow to be resilient when they aren't. These are directly transferrable to infrastructure planning for scale-up.¹⁶⁸
 - **"Offline-first" content delivery and solar power where grids are weak.** Emerging deployments in Nigeria use Kolibri to serve rich, standards-aligned content over local networks with intermittent or no Internet, while a recent Nigerian study documents how solar-powered Wi-Fi hubs can bridge access in remote communities. They outline a low-cost stack for AI-readiness: solar, local server (offline LMS) and periodic sync for updates and model/tool improvements, crucial for rural schools and learning centres.¹⁶⁹
 - **Telco/CSR labs as equity accelerators, if tied to upkeep.** Private partners are refreshing ICT/science labs across public schools (e.g., MTN Foundation's 29 labs in NE/SE), creating immediate "on-ramps" for digital and AI use. Best practice is to secure service contracts and consumables (data, device refresh, minor repairs) in MOUs so labs don't decay after ribbon-cutting; align upgrades with teacher PD and AI-safe use policies.¹⁷⁰
 - **Inclusive community access through adult-learning hubs.** UNESCO's digital-literacy pilots in Nigeria (e.g., Cross River) show how community centres can extend school infrastructure after hours for adults and out-of-school youth. As AI enters classrooms, co-locating devices/connectivity with non-formal programmes helps entire households build readiness (parents, caregivers), improving take-up and safeguarding.¹⁷¹
 - **Pipeline scale concepts to watch.** Large device/connectivity proposals (e.g., the reported UBEC partnership to distribute laptops/tablets nationwide) could dramatically expand access if they bake in equity filters (girls/rural/low-income prioritization), service-level guarantees, and offline-first content.¹⁷² These should be paired with teacher support and school-level maintenance budgets to convert hardware into sustained learning capacity.
- What these cases suggest for policy design: blend USPF-style funding for last-mile connectivity; require offline-first, solar in low-grid areas; link CSR/telco labs to multi-year O&M and teacher PD; use community hubs to widen access beyond enrolled pupils; and make every large-scale device initiative service-backed and equity-targeted. Combined, these moves turn sporadic projects into an infrastructure fabric that lets AI tools work reliably for all learners, not just the best-connected schools.

¹⁶⁷ https://uspf.gov.ng/images/USPF_AR_2018_Final_Draft.pdf

¹⁶⁸ <https://documents1.worldbank.org/curated/en/099548105192529324/pdf/IDU-c09f40d8-9ff8-42dc-b315-591157499be7.pdf>

¹⁶⁹ <https://www.cni.org/topics/digital-curation/scaling-digital-literacy-in-nigeria-kolibris-journey>

¹⁷⁰ <https://www.mtn.ng/newsfoundation/mtn-foundation-completes-the-renovation-of-29-laboratories-across-nigerias-northeast-and-southeast/>

¹⁷¹ <https://www.unesco.org/en/articles/how-digital-learning-improving-livelihoods-nigeria>

¹⁷² <https://www.arise.tv/ubec-us-firm-to-roll-out-47-million-laptops-tablets-across-nigeria-in-landmark-digital-education-project/>

TABLE 5 Cross-cutting Lessons for Nigeria

What Works	Why It Matters
Offline/low-bandwidth AI deployment	Makes adoption possible in rural and low-connectivity schools
Solar power & school-based device access	Addresses energy & device scarcity hindering continuity
Teacher-led facilitation	Supports responsible use, trust, and classroom effectiveness
Gender-targeted inclusion models	Prevents AI from amplifying existing disparities
Public–private partnerships	Unlock scale beyond donor pilots

Strategic Infrastructure & Access Strategies

To ensure reliable access for AI-enabled learning, infrastructure should be planned as a service ecosystem rather than one-off hardware deployment. This means securing multi-year maintenance, power stabilization, and bandwidth guarantees in all school connectivity programs, supported by contractual service-level agreements and public–private procurement standards. Such approaches are aligned with ongoing national initiatives aimed at expanding school connectivity and digital modernization.

A tiered infrastructure model should be adopted to ensure that schools in low-grid or high-poverty locations are not left behind:

- **Tier 1** (Urban & peri-urban): Fiber or high-bandwidth school connectivity, AI-capable device pools, and smart labs.
- **Tier 2** (Rural): Solar-backed Wi-Fi hotspots and school-cluster connectivity for offline-first AI content.
- **Tier 3** (Hard-to-reach): Community digital-learning hubs that serve schools and adult learners together, leveraging satellite or periodic sync.

This aligns with best practices from digital-skills pilots supporting underserved populations.

Strategic partnerships should leverage telecoms, energy innovators, and local EdTech to scale equitable access. Telco CSR initiatives already upgrading school labs must transition into co-financed footprint expansion tied to AI-learning goals, while innovations such as solar-powered connectivity points provide cost-effective reach into remote areas. Public investment should be focused on expanding reach into underserved regions, while private partners deliver cost efficiencies in device supply and last-mile solutions.

Every infrastructure intervention must include a safety and access equity lens: safeguarded accounts for girls, learners with disabilities, and marginalized groups; affordable data solutions; and secure device-sharing protocols. This aligns with global warnings that lack of safe access can widen gender and socioeconomic digital divides as AI expands into learning environments.

Finally, infrastructure investments should be sequenced with teacher readiness and content availability, so that connectivity translates immediately into instructional value. The Edo State AI pilot demonstrated that even when infrastructure is imperfect, structured timing and supported delivery can yield strong learning gains, a principle that must guide phased national scale-up.



AI Content and Resource Development

Current Situation Analysis / Gaps: Guidelines for culturally relevant, multilingual AI educational content are urgently needed, as Nigeria's more than 500 languages contrast sharply with the current reality where most AI-enabled learning materials remain English-centric and imported from Western contexts. For instance, the launch of Nigeria's "N-ATLAS" language model emphasises the need to "safeguard Nigeria's cultural and linguistic diversity" by building local language AI resources.¹⁷³ A commentary on digital inclusion notes: "one of the most persistent barriers ... is not merely access to technology, it is language. ... AI presents a transformative opportunity to redefine education technology in a way that is not only inclusive but also culturally and contextually relevant."¹⁷⁴

Open educational resources and AI-powered adaptive learning platforms remain limited in Nigeria, as review of AI-research in Nigeria found that while AI technologies (e.g., student-performance prediction, multimedia e-learning systems) are increasingly referenced, sophisticated systems like intelligent tutoring systems or large-scale adaptive platforms remain rare.¹⁷⁵ For example, in TVET institutions the use of AI-driven adaptive learning systems is marked as "a unique opportunity ... but implementation ... faces several challenges such as inadequate infrastructure, limited access to technology, and insufficient training for educators and learners."¹⁷⁶

Collaboration with academia, civil society, and technology developers is essential because research emphasises the need for localized content creation, "personalised learning, driven by AI, adapts content to individual learner profiles ... accessibility innovations such as multilingual translation and voice-enabled apps address Nigeria's vast linguistic diversity."¹⁷⁷ That suggests that content development must involve local educators, linguists, developers and community actors, yet institutionalised collaboration mechanisms are still weak.

Data protection and privacy protocols in AI content management require major strengthening in Nigeria because studies indicate concerns around ethical use: for example, in Nigerian TVET institutions one of the barriers to adoption cited is "data privacy and security" in AI-driven systems.¹⁷⁸ Also, global content-development discussions note that AI-generated educational content quality depends on "contextual relevance across five levels: language, region, community, classroom, and individual child."¹⁷⁹ Without hygiene around data governance, algorithmic bias, or equity of access, AI content risks perpetuating inequities.

Also, producer capacity and market signals are weak. Universities, publishers, edtech startups and teacher communities lack shared standards, commissioning pathways, and incentives to produce AI-ready OER (with structured metadata, datasets, and formative assessments) that ministries can adopt at scale.¹⁸⁰ Opinion and sector pieces in Nigeria underline the opportunity for multilingual, mobile-first content to expand digital skills but turning that into repeatable, quality-assured AI content requires coordinated funding, templates, and open licensing.

Statistics and indicators reveal that while exact national stats on AI-content for Nigerian education are limited, research shows from 2008-2022 that 74 Nigerian-based articles studied AI in education, but many focused on basic e-learning systems rather than advanced adaptive content.¹⁸¹ This highlights a gap between available content and the high-quality adaptive, localized AI resources required for scale.

Case Study / Best Practice: One recent international analysis of AI-content generation and curation in education indicated two distinct design approaches: (i) large adaptive-model systems that personalise existing high-quality content (e.g., Squirrel AI), and (ii) teacher-empowerment tools that give educators AI-workbenches to generate culturally contextualised content (e.g., Chimple).¹⁸²

¹⁷³ <https://fmcide.gov.ng/nigeria-launches-landmark-ai-model-powered-by-awarri-to-advance-languages-and-ai-at-scale/>

¹⁷⁴ <https://isnhubs.org.ng/2025/04/24/multilingual-learning-in-nigeria/>

¹⁷⁵ <https://link.springer.com/article/10.1007/s44163-024-00163-y>

¹⁷⁶ <https://www.iardjournals.org/get/WJIMT/VOL.%208%20NO.%205%202024/Exploring%20AI-Driven%20Adaptive%20188-198.pdf>

¹⁷⁷ <https://rijessu.com/wp-content/uploads/2025/09/007-RIJE-2025-V4-NO4-009.pdf>

¹⁷⁸ <https://www.iardjournals.org/get/WJIMT/VOL.%208%20NO.%205%202024/Exploring%20AI-Driven%20Adaptive%20188-198.pdf>

¹⁷⁹ <https://edtechhub.org/2025/07/23/how-are-education-implementers-approaching-ai-content-generation-curation-and-integration/>

¹⁸⁰ <https://isnhubs.org.ng/2025/04/24/multilingual-learning-in-nigeria/>

¹⁸¹ <https://link.springer.com/article/10.1007/s44163-024-00163-y>

¹⁸² <https://edtechhub.org/2025/07/23/how-are-education-implementers-approaching-ai-content-generation-curation-and-integration/>

In Nigeria, the N-ATLAS initiative (see above) exemplifies efforts to build multilingual / culturally relevant AI content at national scale.¹⁸³

Additionally, the use of AI-driven adaptive learning systems in Nigerian TVET institutions has been documented as promising: “AI-driven adaptive learning systems have the potential ... equipping students with relevant skills and enhancing their engagement and motivation.”¹⁸⁴ and should involve partnerships across sectors (education, technology, language communities).

A study titled “The Role of Artificial Intelligence in Enhancing Adult Education in Nigeria” (Adeyinka et al., 2025) explores how AI-driven systems such as virtual tutors, chatbots, and adaptive learning platforms can be integrated into adult education programmes across Nigeria.¹⁸⁵ It notes that although infrastructure and resources remain significant barriers, early pilots show the potential for facilitating personalized learning paths for adult

learners with diverse needs and schedules—, highlighting improved access when learning is tailored to the learner rather than the other way around.

Another emerging example is adult-learning centres in Nigeria (for instance in Jigawa State) that have begun trial-deploying AI-powered adaptive literacy platforms to accelerate reading and numeracy among adult learners. One case study indicates that AI tools enabled real-time diagnostics of learner difficulties and provided targeted remediation, reducing drop-out rates and boosting engagement.¹⁸⁶ While scale remains small and formal evaluation limited, these pilots illustrate how content localisation (language, context), micro-learning modules and learner-paced platforms can significantly improve the relevance and reach of AI content in non-formal settings.



¹⁸³ <https://fmclide.gov.ng/nigeria-launches-landmark-ai-model-powered-by-awarri-to-advance-languages-and-ai-at-scale/>

¹⁸⁴ <https://www.iiardjournals.org/get/WJIMT/VOL.%208%20NO.%205%202024/Exploring%20AI-Driven%20Adaptive%20188-198.pdf>

¹⁸⁵

https://www.researchgate.net/publication/388707917_THE_ROLE_OF_ARTIFICIAL_INTELLIGENCE_IN_ENHANCING_ADULT_EDUCATION_IN_NIGERIA_Technology_Education_under_the_Theme_%27%27Innovative_Technology_and_Pedagogy_for_Best_Practice_and_Sustainable_Development_in_N

¹⁸⁶

<https://www.dataprojectng.com/project/12577/An%20Assessment%20of%20the%20Applications%20of%20Artificial%20Intelligence%20in%20Literacy%20Improvement%20Programs%203A%20A%20Case%20Study%20of%20Adult%20Education%20Centers%20in%20Jigawa%20State>

Key Recommendations for Improvement

Establish national standards for AI-ready content: Publish “Nigeria AI-in-Education Content Standards” covering pedagogy, accessibility (WCAG), multilingual/localisation, dataset provenance, bias checks, and age-appropriateness. Require structured metadata (subject, grade, language, disability supports, alignment to NERDC curricula) so resources are discoverable and interoperable.

Build a localisation pipeline (languages & culture): Fund rapid translation/adaptation of core materials into the top Nigerian languages, with community review panels. Incentivise locally relevant case studies (agriculture, health, climate, entrepreneurship) and indigenous knowledge modules.

Launch a national AI-OER repository: Create a government-hosted, API-accessible library for AI-ready OER (lesson objects, quizzes, datasets, prompts), licensed openly for reuse. Add educator toolkits: prompt libraries, ethics mini-lessons, assessment rubrics, and offline packages for low-connectivity areas.

Prioritise adaptive and offline-first delivery: Approve a shortlist of adaptive engines (including offline/sync options) that run on low-spec devices. Package content for offline hubs (school clusters, community centres) with periodic sync; include solar-ready deployment guides.

Protect learners’ data and rights: Enforce a Student Data Protection Code for content platforms: consent, minimal data collection, local storage where feasible, deletion timelines, audit trails, and child-online safety controls. Mandate model/content bias and safety reports before procurement or scale-up.

Co-create with teachers, universities, and startups: Commission content through “creation sprints” pairing teachers, subject experts, local edtech and linguists; pay per accepted, standards-compliant asset. Set up regional Content Studios (teacher fellows + student interns) to keep materials updated and contextual.

Strengthen teacher capacity for AI content use: Embed short, stackable PD micro-credentials: AI pedagogy, prompt design, formative assessment with AI, and data ethics. Create moderated communities of practice for sharing classroom-tested AI lessons and adaptations.

Align TVET and adult education: Develop sector-specific AI modules (agri-analytics, health informatics, logistics, manufacturing) with employer input. Offer mobile-first micro-courses and micro-credentials recognised by industry associations.

Finance for scale and sustainability: Dedicate budget lines for content commissioning, localisation, maintenance and accessibility updates. Use PPPs for platform hosting, with SLAs for uptime, security, language coverage and accessibility features.

Monitor quality and impact: Track KPIs – % of content in Nigerian languages; # of accessible resources; monthly active teacher/learner users; usage in rural schools; learning gains from adaptive modules.

9.

Ethical, Legal, and Governance Framework

Education-Specific AI Ethics Code: AI systems used in education must adhere to a code grounded in ethics, human rights, and trustworthiness. International research emphasises key ethical concerns such as bias, lack of transparency, and threats to student privacy.¹⁸⁷ According to a systematic review: “Data privacy and security ... bias and fairness ... the role of human oversight ... consent and autonomy ... transparency and accountability” are major ethical pillars in AI for education.¹⁸⁸ In alignment with the UNESCO Recommendation on the Ethics of Artificial Intelligence, the deployment of AI “depends on their transparency & explainability (T&E)” and must respect human dignity.¹⁸⁹ Nigeria’s education system need to adopt a sector-specific ethics code that governs the design, procurement, classroom use, and evaluation of AI tools in basic, secondary, TVET, higher, and non-formal education. The code should align with the Nigeria Data Protection Act 2023 (NDPA), which establishes the Nigeria Data Protection Commission and safeguards learners’ personal data, and with the UNESCO Recommendation on the Ethics of AI (2021) which frames human rights, dignity, transparency, fairness, and human oversight as cornerstone principles. It should also reference Nigeria’s draft National AI Strategy (NAIS), which places responsible and inclusive adoption at the centre of sector transformation.¹⁹⁰

Core principles.

- **Equity & non-discrimination:** AI must not exacerbate educational inequities; conduct pre-deployment bias and impact assessments with gender, disability, language, and location lenses. (UNESCO Rec: fairness, inclusion, human rights).¹⁹¹
- **Transparency & explainability:** Students, teachers, and parents should understand what the system does, what data it uses, and how outputs affect grading, placement, or support; provide plain-language model cards and user notices.
- **Human oversight:** Teachers remain accountable for pedagogical judgments; AI augments, not replaces, educators.
- **Privacy & data protection:** Processing of learner data must follow NDPA lawful bases, data minimization, purpose limitation, children’s data safeguards, security controls, and defined retention/deletion schedules; high-risk use requires a Data Protection Impact Assessment (DPIA) and NDPC-compliant governance.¹⁹²
- **Safety & integrity:** Guard against hallucinations, harmful content, academic dishonesty, and over-automation; publish acceptable-use norms for teachers and learners.

¹⁸⁷ <https://www.enrollify.org/blog/ethical-considerations-for-ai-use-in-education> xd

¹⁸⁸ <https://thecasehq.com/ethical-considerations-of-ai-in-education-balancing-innovation-and-privacy/>

¹⁸⁹ <https://www.unesco.org/en/artificial-intelligence/recommendation-ethics>

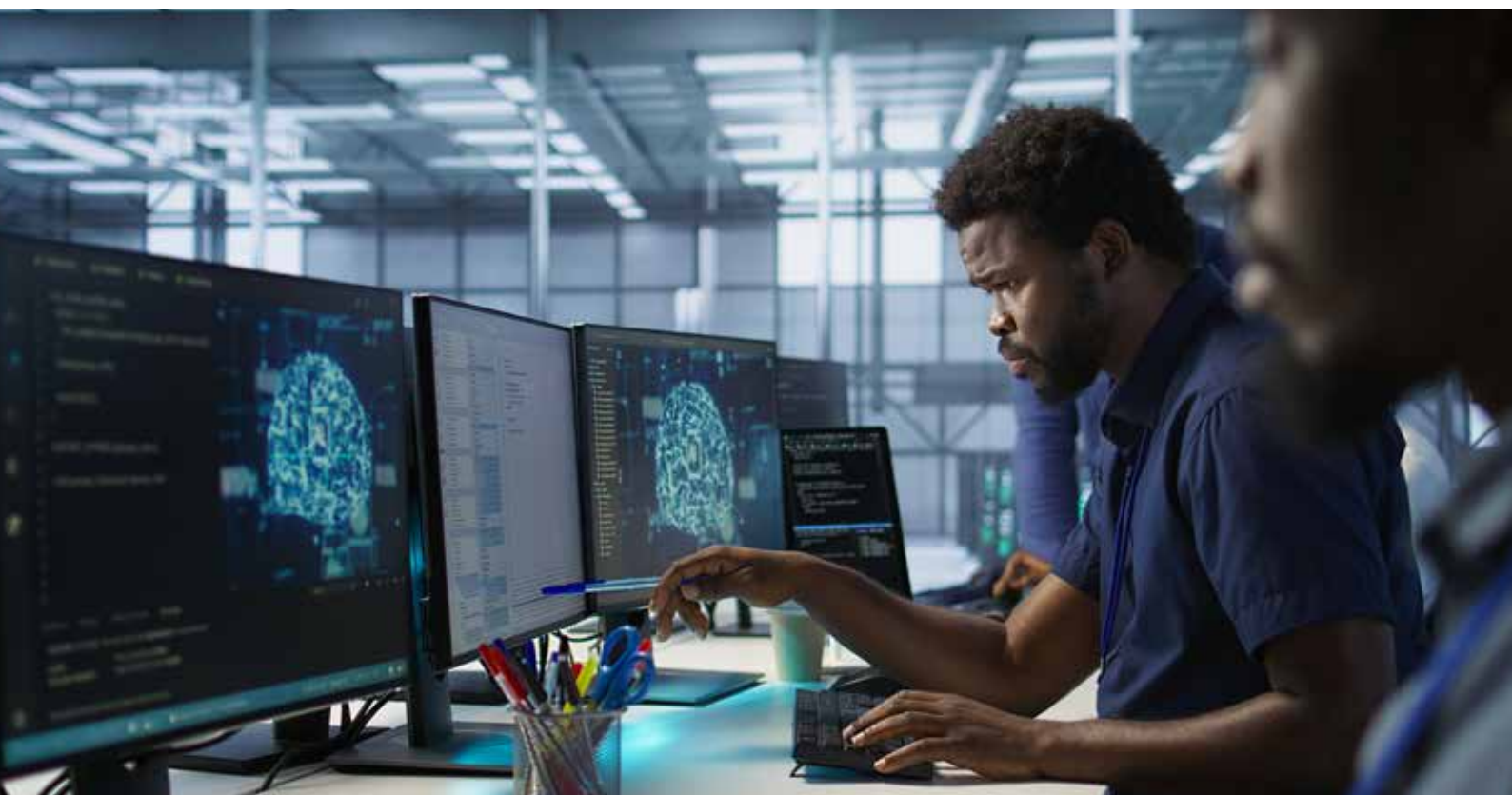
¹⁹⁰ https://cert.gov.ng/ngcert/resources/Nigeria_Data_Protection_Act_2023.pdf

¹⁹¹ <https://www.unesco.org/en/artificial-intelligence/recommendation-ethics>

¹⁹² https://cert.gov.ng/ngcert/resources/Nigeria_Data_Protection_Act_2023.pdf

Operational requirements for schools and providers.

1. Data governance: Maintain an AI register of tools in use; document purposes, datasets, retention, vendors, and DPIA outcomes; ensure parental/guardian notices and age-appropriate consent for minors under NDPA.
2. Model quality & bias checks: Require vendors to disclose training data provenance, evaluation methods, known limitations, and mitigation steps; run periodic fairness audits on learning and placement outcomes.
3. Transparency to learners: Provide in-product disclosures ("AI is assisting this feedback"), opt-out alternatives where feasible, and clear redress routes to a school AI lead and the NDPC for privacy complaints. (NDPA/NDPC).¹⁹³
4. Accountability & audit: Set school-level AI Leads and state/national oversight that can suspend a tool if risks materialize; publish annual transparency reports. Note that NDPC has begun active enforcement in Nigeria's data ecosystem underscoring the need for compliance-ready practices in education.¹⁹⁴
5. Accessibility & inclusion by design: Mandate multilingual interfaces, assistive-tech compatibility, low-bandwidth modes, and offline-first options to avoid excluding rural schools and learners with disabilities.
6. Teacher empowerment & safeguards: Require teacher training on ethical AI use, prompt design, bias recognition, and safe classroom practices; define boundaries (e.g., no AI-only grading of high-stakes assessments). (UNESCO Rec).¹⁹⁵



¹⁹³ <https://ndpc.gov.ng/>

¹⁹⁴ <https://www.reuters.com/business/finance/nigerian-data-agency-fines-fidelity-bank-breaches-2024-08-22/>

¹⁹⁵ <https://www.unesco.org/en/artificial-intelligence/recommendation-ethics>

Roles and Responsibilities of Institutions & Regulators

Effective governance of AI in education requires clearly defined responsibilities across federal agencies, state authorities, educational institutions, private developers, and the community. Regulators and government bodies lead on national policy direction, creating the enabling frameworks that ensure AI serves education goals safely and inclusively. This involves establishing and updating standards on data protection, cybersecurity, accessibility, and algorithmic fairness, as well as setting certification and approval processes for AI tools used in Nigerian classrooms. Oversight agencies must also monitor compliance, conduct audits, and issue sanctions where student rights or ethical obligations are breached, ensuring that responsible AI is not optional but a system requirement.

Educational institutions have a frontline role in translating these regulations into safe and effective classroom practice. Schools, colleges, and universities determine which AI tools to procure and how to integrate them into curricula, pedagogy, and assessment while upholding human oversight by trained educators. They are responsible for providing ongoing teacher professional development in ethical AI use, bias recognition, and digital-safety practices. Institutions must also maintain usage records, conduct outcomes-monitoring, protect student data, and establish clear channels for reporting concerns and redress, ensuring accountability reaches the point of implementation.

Private sector actors particularly EdTech companies play an essential role as technology providers but must comply with ethical and safety expectations. They should embed transparency features such as explainable AI, publish model limitations, enable accessibility in low-bandwidth settings, and provide audit access for regulators and schools. Collaboration with educators is crucial to ensure contextualization, local languages, curriculum alignment, relevance for Nigerian labour-market needs—and to avoid the importation of bias from foreign datasets or design

assumptions.

Finally, **learners, parents, and communities** must be empowered as active stakeholders. They require clear communication on where and how AI is used, what data it collects, and how decisions affecting learning are made. Consent processes should be age-appropriate, and families must have channels to provide feedback or contest harmful outcomes. Building digital trust is especially important for girls and vulnerable learners who may face disproportionate online risks. As one empirical study cautions, “inconsistent regulatory approaches contribute to disparities in AI deployment ... bias mitigation strategies, explainable AI mechanisms, and robust data governance policies are needed” to ensure AI integration does not deepen inequality but actively promotes safe, equitable learning opportunities across Nigeria.

Transparency, Accountability, and Stakeholder Engagement Processes

Transparency and accountability are central to trusted AI systems in education. Research shows that when decision-making processes are opaque, trust is undermined.¹⁹⁶ Educational institutions should publish clear information on how AI tools work (for example, how data are used, what criteria algorithms apply, what recourse exists for errors).¹⁹⁷ Stakeholder engagement, including teachers, students, parents, communities, must be ongoing: there should be mechanisms for input, feedback and remediation when AI systems produce adverse outcomes. Audits, independent reviews and reporting frameworks help ensure accountability and support corrective action.

Risk Management: AI Misuse, Data Privacy & Bias

Deploying artificial intelligence in educational settings brings distinct risks that must be actively managed. Among these are misuse of AI systems, such as surveillance, profiling, or decision-making with insufficient human oversight, and breaches of data privacy when learner data are collected, stored, used or shared without appropriate safeguards. Research into AI in education shows that algorithmic bias and inequitable access

¹⁹⁶ <https://www.frontiersin.org/journals/human-dynamics/articles/10.3389/fhumd.2024.1421273/full>

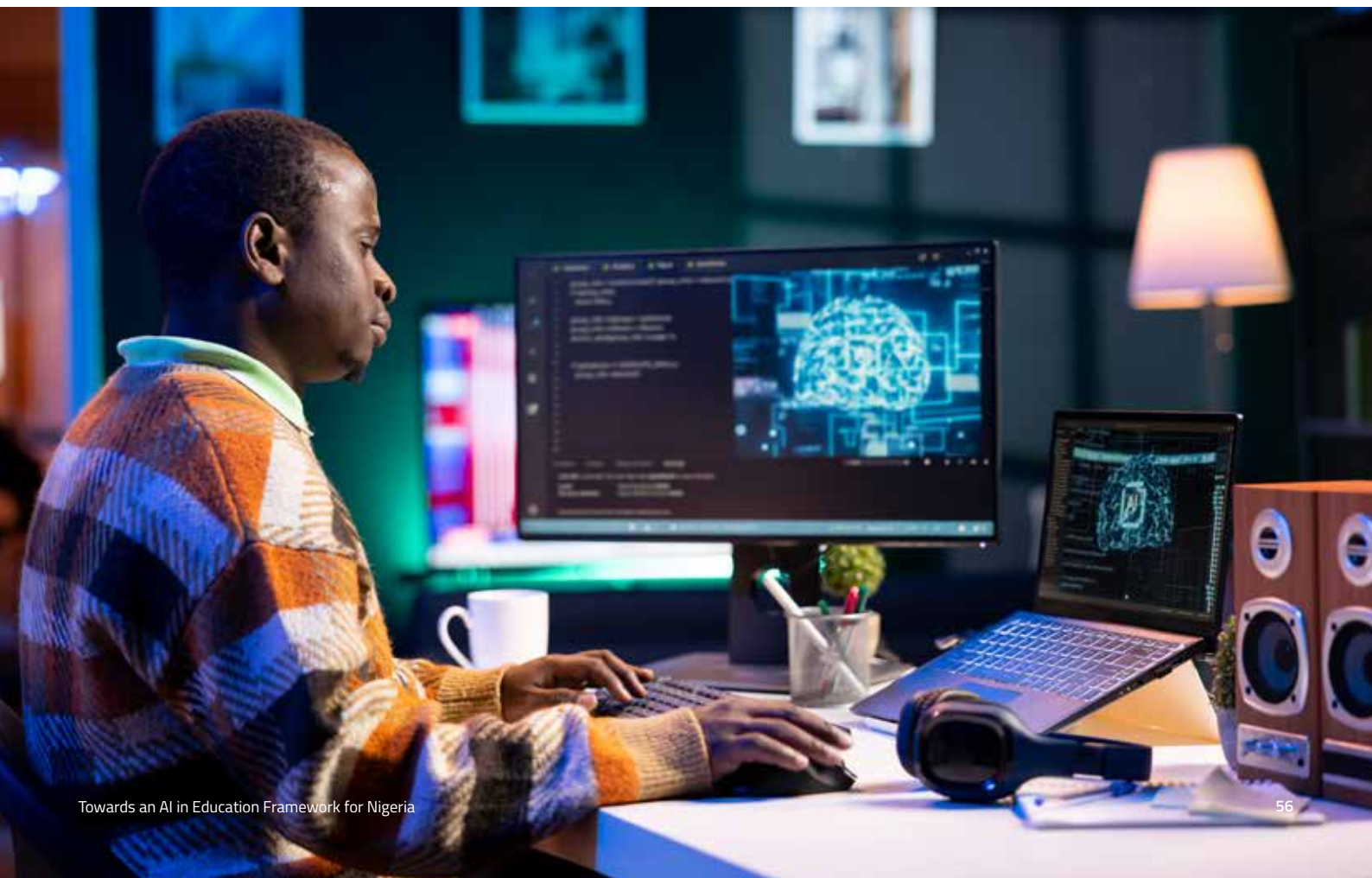
¹⁹⁷ <https://genai.calstate.edu/communities/faculty/ethical-and-responsible-use-ai/ethical-principles-ai-framework-higher-education>

¹⁹⁸ https://www.researchgate.net/publication/364325080_Ethical_principles_for_artificial_intelligence_in_education

remain major obstacles, with certain student groups at risk of disadvantage if AI systems replicate or amplify existing inequalities. For instance, one review observes that “algorithmic fairness and equitable access are still major ethical issues in AIED.”¹⁹⁸ Without robust risk management, AI systems could contribute to deskilling of teachers, reduction of human judgment, unintended harms to learners, and loss of trust in educational institutions.

To mitigate these risks, a comprehensive risk-management strategy is required.

- First, pre-deployment impact assessments should evaluate potential harms using fairness, inclusion, and bias lenses to flag unintended consequences before a tool is scaled.
- Second, data-protection protocols must govern every stage of the data lifecycle: from informed consent and anonymisation, through secure storage and defined deletion schedules, to transparency about how data will be used.
- Third, institutions must establish mechanisms for error/harms detection and redress for example giving students and caregivers the right to appeal decisions made by AI systems, correct data or challenge outcomes.
- Fourth, periodic auditing and monitoring of deployed AI tools, bias audits, transparency reports, outcome reviews, are essential so the system remains accountable, adaptive and equitable over time.
- Finally, capacity-building and awareness for teachers, administrators and technology providers are critical: ethical AI use must be built into professional development so that these stakeholders understand their roles, limitations of AI, and how to intervene responsibly when systems go off track.



10.

Monitoring, Evaluation, and Research

Current Situation Analysis / Gaps: Nigeria's education system is still in the early stages of establishing a robust monitoring, evaluation, and research (MER) architecture for AI-enabled learning. While digital tools can generate rich data streams, clear and context-appropriate metrics to evaluate AI implementation and actual learning outcomes remain underdeveloped. Evidence from higher education shows that even where KPIs exist, institutions face challenges around data reliability, stakeholder resistance, and alignment with diverse institutional goals, making it difficult to assess whether AI improves learning, supports teachers, or simply adds technical complexity.¹⁹⁹ At the school level, existing performance monitoring frameworks rarely include AI-specific indicators such as digital engagement, adaptive-task completion, accessibility for marginalized learners, or ethical-use compliance.²⁰⁰

Similarly, feedback loops and data-driven decision-making are not yet embedded in most institutions adopting AI. Although AI dashboards can theoretically provide real-time insights, attendance, skill progression, and content mastery systems to translate these signals into actionable instructional or policy responses remain weak.²⁰¹ Without structured cycles of analysis, teacher

support, and intervention, the learning gains observed in controlled pilots risk being lost in day-to-day delivery.

There is also limited systematic research on AI's pedagogical impact in Nigeria. Global reviews point to both the promise and risks of AI-enabled assessment including "automated scoring, rapid content analysis and personalised feedback" but warn that validity, reliability, transparency, fairness and equity must be rigorously studied.²⁰² In Nigeria, evidence is still scarce outside a few notable pilots. Little is known about how AI adoption affects gender equity, rural inclusion, teacher workload, or long-term learner motivation. This leaves decision-makers with insufficient evidence to guide scale-up or course-correction.

Moreso, policy learning cycles remain infrequent. Most education policy structures are static and not designed to respond to rapidly evolving technologies. Without ongoing evaluation and periodic recalibration, AI deployments risk becoming outdated, inequitable, or misaligned with national education priorities. A resilient MER system must therefore ensure continuous insight generation—feeding real-time evidence into timely updates to curricula, procurement rules, capacity-building, and governance frameworks.



¹⁹⁹ "data reliability, stakeholder resistance, and the difficulty of aligning KPIs with diverse institutional goals." ResearchGate

²⁰⁰ <https://www.hurix.com/blogs/four-proven-metrics-to-assess-ais-impact-on-curriculum-design/>

²⁰¹ <https://schoolai.com/blog/how-ai-helps-school-admins-optimize-performance-monitoring/>

²⁰² <https://arxiv.org/abs/2406.18900>

Proposed Monitoring & Evaluation Framework

Track whether AI in education is implemented with quality, improves learning, and advances equity, while safeguarding learners and informing iterative policy updates.

A. Implementation & Readiness

Indicator	What it Measures	Data Source	Frequency	Disaggregation
% schools with minimum AI-readiness (power, connectivity, device: learner ratio, approved tools)	Basic implementation capacity	EMIS/School census; infrastructure audits	Termly/Annual	State/LGA, urban–rural, school type
% schools with an AI tool register & DPIA completed	Governance readiness	School compliance reports; regulator spot-checks	Termly	State/LGA, ownership
% teachers trained to baseline AI pedagogy & ethics standard	Human capacity	TPD records; certification logs	Termly	Phase (JSS/SSS/TVET), gender

B. Usage & Engagement

Indicator	What it Measures	Data Source	Frequency	Disaggregation
Active use rate of AI platforms (weekly active users/eligible)	Adoption at classroom level	Platform logs (privacy-preserving)	Monthly	Grade, subject, gender
Time-on-task & completion of adaptive activities	Depth/quality of use	Platform analytics	Monthly	Grade, subject, school
Teacher uptake of AI-assisted lesson prep/feedback	Pedagogical integration	Teacher diaries; LMS logs	Termly	Phase, subject

C. Learning Outcomes

Indicator	What it Measures	Data Source	Frequency	Disaggregation
Δ in subject scores for AI users vs matched non-users (effect size)	Impact on learning	Standardized tests; CA/BECE/WAEC where applicable	Termly/Annual	Gender, location, poverty quintile
Mastery progression in AI/digital competencies	Skills development	Competency-based assessments; micro-badges	Termly	Grade, phase
LAYS gained per learner attributable to AI	Learning efficiency	Evaluation studies (RCT/QED)	Annual	Subgroups (girls, rural, disability)

D. Equity & Inclusion

Indicator	What it Measures	Data Source	Frequency	Disaggregation
Participation gap (AI users %) by gender/rural/SES/disability	Access equity	EMIS + platform logs	Monthly/Termly	Gender, rural/urban, disability
Performance gap closure between priority groups	Outcome equity	Assessments (with controls)	Termly/Annual	Same as above
% content available in local languages & accessible formats	Inclusive content supply	Content audits	Termly	Language, disability

E. Ethics, Privacy & Safety

Indicator	What it Measures	Data Source	Frequency	Disaggregation
% tools with published model cards & data notices	Transparency	Vendor disclosures; school sites	Termly	Vendor, tool type
Reported incidents (privacy breaches, harmful outputs, TF-GBV online) resolved within SLA	Risk management effectiveness	Incident register; helpdesk	Quarterly	Category, severity
Schools passing annual AI ethics & NDPA compliance audit	Compliance quality	Regulator audits	Annual	State/LGA

F. Infrastructure Reliability & Continuity

Indicator	What it Measures	Data Source	Frequency	Disaggregation
Instructional uptime (% sessions not disrupted by power/Internet)	Learning continuity	School logs; IoT/power meters (where present)	Monthly	State/LGA, urban–rural
Device availability (functional devices per 10 learners)	Hardware sufficiency	Inventory audits	Termly	School type
Mean time to repair/replace devices	Maintenance efficiency	Service tickets	Quarterly	Vendor/region

G. Cost & Efficiency

Indicator	What it Measures	Data Source	Frequency	Disaggregation
Cost per learner reached (capex + opex)	Financial efficiency	Budget & PFM data	Annual	Programme, region
Cost per 0.1 SD learning gain	Cost-effectiveness	Evaluation synthesis	Annual	Subgroups
% of budget spent on maintenance & teacher support (vs one-off hardware)	Sustainability orientation	Budget execution reports	Annual	Agency/State

H. User Experience & Well-Being

Indicator	What it Measures	Data Source	Frequency	Disaggregation
Teacher satisfaction/comfort using AI (Likert index)	Teacher experience	PD surveys; focus groups	Termly	Phase, gender
Student perceived usefulness & safety online	Learner experience	Student surveys; sentinel schools	Termly	Gender, age
Academic integrity incidents linked to AI	Safe/ethical use	School discipline logs	Termly	Level, subject

I. Research & Innovation Uptake

Indicator	What it Measures	Data Source	Frequency	Disaggregation
# rigorous studies (RCT/QED) completed and publicly shared	Evidence generation	Research registry	Annual	State/topic
% policy or programme changes citing MER evidence	Evidence-to-policy	Policy memos; MoE minutes	Annual	Sector (basic/TVET/HE)
# locally-developed AI content/tools adopted system-wide	Local innovation scaling	Procurement records	Annual	Tool type

Data Governance & Feedback Loops (How to Use the Indicators)

- Privacy-first analytics: Collect only necessary, de-identified usage metrics; apply NDPA-compliant DPIAs for high-risk analytics.
- Dashboards at three levels: school (instructional adjustments), state (resource targeting), national (policy steering).
- Review cadence: monthly operational reviews (usage/infrastructure), termly pedagogical reviews (learning/equity), annual public report (ethics/compliance, cost-effectiveness).
- Adaptive management: when indicators fall below thresholds (e.g., girls' participation < 45% of users; uptime < 85%), trigger corrective actions additional devices, data subsidies, targeted PD, or vendor remediation.

Research & Innovation Agenda

Nigeria should prioritize pilot-and-scale studies that test AI in real classrooms and non-formal settings—especially where resources are scarce. High-value pilots include adaptive tutoring in low-infrastructure schools, generative AI for multilingual African languages (to widen access and relevance), and AI-supported assessment for remote or displaced learners. Each pilot should be designed with scale in mind (clear cost models, equity safeguards, and teacher enablement) so that proven approaches move from small trials to statewide or national adoption without losing fidelity.

To generate credible evidence, build institutional research capacity across universities, colleges of education, and independent research centres. Encourage rigorous methods randomized controlled trials, quasi-experimental designs, and mixed-methods studies to understand not only what works but for whom and under what conditions. Parallel to this, launch privacy-preserving open-data initiatives so de-identified learner and system data can fuel

machine-learning models, equity analytics, and policy dashboards. Strong data governance (consent, minimization, DPIAs) is essential to protect learners while enabling insight.

There is need to foster an innovation ecosystem that connects edtech start-ups, teacher-innovators, NGOs, and policymakers. Create rapid research-to-practice loops, e.g., challenge funds and living labs so findings directly inform product design, curriculum alignment, and procurement. Make ethical-AI research a first-class priority: study bias and fairness in educational models, data-governance in schools, transparency and explainability in student-facing tools, and the effects of AI on teacher–learner relationships. This agenda will ensure AI adoption advances learning, equity, and trust—grounded in Nigerian evidence and capabilities.

On institutional research capacity, open data, innovation, and ethical-AI studies, there are still important gaps to address. We need more long-term research to understand how AI's impact on learning and teaching evolves over time, not only during short pilots. There is also a need for implementation research that examines what makes schools succeed or struggle when adopting AI including leadership, teacher training, budgeting, and community acceptance. In addition, cost-effectiveness studies are needed so that decisions about scaling are grounded in affordability and sustained financing, not temporary donor support.

Another gap is the connection between AI learning and economic opportunities. More research should look at how AI-related skills feed into recognized certificates, employability, and sector-specific skills needs in Nigeria. We also need more studies focused on learners with disabilities, displaced learners, and rural adults in groups who risk being left behind in AI adoption. Finally, there is room to expand research on how AI affects teaching workload, academic integrity, student motivation, and mental health. Addressing these missing areas will ensure that AI in education advances learning, equity, and future work outcomes for all learners in Nigeria.

11.

Financing and Partnerships

Current Situation Analysis / Gaps: Government budget allocations specific to AI in education are still limited in Nigeria. While Nigeria has overarching digital economy and human-capital initiatives, targeted budget lines for AI in education remain few and weakly defined. A review of EdTech financing in low- and middle-income countries finds that “the primary source of funding for EdTech in most countries is public funds ... another major source is the Universal Service Access Fund (USAF) ... public-private partnerships (PPPs) are rarer and have not been developed to their full potential.”²⁰³ In Nigeria, a study of AI in school education notes: “the national strategy ... only vaguely mentions initiatives to introduce AI curricula in schools ... lacks clear policy direction for AI education and how to achieve AI education for all.”²⁰⁴

Leveraging private sector investment, donor funds and international aid is becoming increasingly important as Nigeria advances its AI-in-education agenda. There’s growing private-sector interest in AI education in Nigeria, and global trend lines suggest PPPs can play a key role. A Cisco/CMU white paper on Africa highlights that “through private sector innovation ... PPPs can bridge resource constraints to achieve scalable AI solutions tailored to African needs.”²⁰⁵ Similarly, an article on “Leveraging PPPs for African AI growth” states: “public-private partnerships can be crucial in promoting inclusive education and skill development programs focused on AI.”²⁰⁶ However, many donor funds and international aid initiatives are still funnelled into pilots rather than large-scale sustainable AI-in-education programmes.

A framework for sustainable public–private partnerships and community engagement is essential to scale AI in education effectively. Studies of PPPs in Nigerian education (even at primary level) show positive correlation between PPP implementation and improved funding, but also flag challenges: “there is a positive relationship between implementing public private

partnership and improved funding of primary education schools ... but there are significant challenges in accountability, quality control and governance.”²⁰⁷ Thus, while there is potential for PPPs in AI education, a structured national framework guiding them is underdeveloped.

Nigeria’s edtech ecosystem is growing steadily, with startups such as Afrilearn, Edves and 9jaCodeKids highlighting local innovation in K-12 digital learning and STEM programmes.²⁰⁸ Yet, many of these ventures remain siloed and small-scale, with limited pathways into large-scale adoption across public education systems. For sustainable AI-in-education in Nigeria, stronger linkages between local innovators, educational institutions and government financing mechanisms are essential. Though hubs such as Co Creation Hub and Ilorin Innovation Hub offer incubation and mentoring, the absence of dedicated seed funds, innovation challenge funds, or incubation models tied specifically to AI-in-education means localised solutions often lack the growth support needed to move from prototype to wide deployment.²⁰⁹

Many existing investments in educational technology in Nigeria remain project-based or pilot-oriented, rather than rooted in sustained, recurring funding streams. A World Bank working paper on innovative financing of EdTech observes that major public funding often comes from election manifesto promises, stimulus packages and emergency response funds, which are “more temporary or sporadic” compared to routine ministry budget lines.²¹⁰ Without budget lines that explicitly support AI in education, covering content creation, teacher training, device maintenance and data infrastructure there is a risk that AI interventions will not be sustained or will favour better-resourced schools, reinforcing inequities. Observers in the Nigerian context have also called for far greater financing of digital and ed-tech solutions if they are to achieve impact and scale across the country.

²⁰³ <https://thedocs.worldbank.org/en/doc/a607bb6e3b76d2be0f3db8db34dcf73e-0140022025/related/9EDU-WP-19-Innovative-financing-of-education-technology.pdf>

²⁰⁴ <https://journals.openedition.org/ries/17323?lang=es&>

²⁰⁵ https://newsroom.cisco.com/c/dam/r/newsroom/pdfs/Cisco-CMU-Whitepaper_AI-and-the-Workforce-in-Africa.pdf

²⁰⁶ <https://convergenceai.io/leveraging-ppps-for-african-ai-growth/>

²⁰⁷

<https://njeapjournal.com/wp-content/uploads/2024/10/3-Public-Private-Partnership-PPP-Implementation-as-Correlates-for-Improved-Primary-Education-Funding-in-Abia-State-Nigeria.pdf>

²⁰⁸ <https://www.edtechstartups.org/country/Nigeria/>

²⁰⁹ <https://ensun.io/search/edtech/nigeria>

²¹⁰ <https://thedocs.worldbank.org/en/doc/a607bb6e3b76d2be0f3db8db34dcf73e-0140022025/related/9EDU-WP-19-Innovative-financing-of-education-technology.pdf>

"I am incredibly proud of our partnership with the Government of Nigeria and how we are working together to make AI literacy available to young students and civil servants. At the launch of this partnership... the Minister for Innovation, Science and Technology, Nigeria, Chief Uche Nnaji, remarked that 'The launch of the AI Academy marks a bold step toward preparing Nigerians for the future. By equipping our citizens with AI skills, we are laying the foundation for a new generation of innovators, problem-solvers, and global leaders who will shape Nigeria's technological and economic progress.'"

The Rt Hon Patricia Scotland KC &
Chief Uche Nnaji, 2025

Best Practice / Case Studies

Case Study 1: Government–Entrepreneur Collaborative EdTech Investment. A study of West and North African countries among them Nigeria analysed the impact of collaborative investment between governments and entrepreneurial venture firms in EdTech. It finds that such joint efforts can lead to meaningful improvements in educational access and quality when public policy aligns with private innovation and funding.²¹¹ The key takeaway: institutionalising these kinds of partnerships helps convert standalone pilots into sustained programmes rather than one-off initiatives.

Case Study 2: Public–Private Partnership (PPP) in Infrastructure at Tertiary Level (Yaba College of Technology). In Lagos, the partnership model adopted by Yaba College of Technology demonstrates how a tertiary institution leverages PPP arrangements to support infrastructure renewal and revenue generation. The study documents how the institution has turned to private-sector partnerships for internal resource

generation and equipment upgrade, thereby easing fiscal pressure.²¹² For AI-in-education financing, this model shows how linking private investment to institutional budgets and capacity strengthening can support sustainability.

Case Study 3: EdTech Financing Ecosystem & Start-Up Growth (Nigeria). An analysis of the EdTech investment landscape in Africa shows that Nigerian start-ups face strong growth potential but also significant financing constraints: in 2023, EdTech received only about 1.4% of all venture capital in Africa.²¹³ These findings highlight the need for innovative financing models (impact investment, challenge funds, blended finance) and demonstrate that showcasing successful case studies can attract more private-sector capital into AI-education initiatives in Nigeria.

Case Study 4: Blended finance to scale EdTech/AI. A World Bank working paper on innovative financing documents country examples where public budget lines were blended with donor funds and PPPs to move beyond pilots and scale digital learning, including AI-capable systems. It outlines practical levers such as better procurement, use of Universal Service/Access Funds, and mobilizing philanthropic and private capital useful design cues for Nigeria's AI-in-education financing architecture.²¹⁴

Case Study 5: PPPs to close infrastructure and capacity gaps (Africa). Continental analyses underscore that PPPs are instrumental in tackling connectivity, power and technical-capacity constraints for AI adoption: "Public-private partnerships are instrumental in addressing infrastructure challenges ... private entities often contribute financial resources, technical expertise, and experience in infrastructure development, while public institutions provide regulatory frameworks ..." ²¹⁵ This division of roles is a transferable template for school connectivity, device supply and shared labs geared to AI learning.

²¹¹ <https://www.syncsci.com/journal/AERE/article/download/AERE.2023.01.003/844/>

²¹² <https://sciencepublishinggroup.com/article/10.11648/j.eas.20220703.12>

²¹³ <https://www.techinafrica.com/challenges-african-edtech-startups-face-in-mobile-learning/>

²¹⁴ <https://thedocs.worldbank.org/en/doc/a607bb6e3b76d2be0f3db8db34dcf73e-0140022025/related/9EDU-WP-19-Innovative-financing-of-education-technology.pdf>

²¹⁵ <https://convergenceai.io/leveraging-ppps-for-african-ai-growth/>

Case Study 6: Nigeria evidence: PPPs correlate with improved funding. A study from Abia State finds a significant relationship between PPP implementation and improved primary-education funding, indicating that the model is viable domestically but requires clear governance, accountability and alignment to learning outcomes when adapted for AI-in-education. Complementary Nigerian studies reach similar conclusions on PPPs for tertiary/secondary infrastructure.

Strategic Recommendations

Building sustainable financing and partnership mechanisms for AI in education requires deliberate policy choices that strengthen local capacity, reduce reliance on short-lived pilots, and secure equitable access nationwide. The following strategic recommendations will ensure long-term resilience, scalability and value for money.

Establish dedicated budget lines for AI in education: Federal and state governments should allocate specific and recurring funds for AI integration, covering digital infrastructure, AI-ready content development, teacher upskilling, device maintenance and responsible-AI governance. This will enable predictable planning and prevent promising AI initiatives from ending once donor funding ends.

Deploy blended financing and structured PPPs: Government should define clear partnership

models that combine public investment with private-sector capital and donor resources. PPP frameworks must include governance mechanisms performance standards, transparency rules, and equity commitments, so that efficiency gains do not come at the expense of access for underserved schools.

Strengthen local innovation and industry linkages: Support Nigerian edtech innovators through seed-funding windows, innovation challenge funds, and incubation programmes focused on AI-enabled learning solutions tied to national curriculum and language contexts. Incentivise partnerships between innovators, universities and teacher-training institutions to speed research-to-deployment cycles.

Ensure financing mechanisms close equity gaps: Funding models should prioritize rural schools, girls, displaced learners, and learners with disabilities. Subsidies for connectivity, community digital hubs, and scalable device-access schemes are vital to prevent AI adoption from reinforcing existing inequalities.

Institutionalize monitoring, accountability & results-based financing: All financing and partnership agreements should require measurable learning outcomes, equity indicators, responsible-AI compliance, and public reporting. Performance-linked disbursement can ensure that investment aligns with student benefits rather than hardware counts alone.

12.

Implementation Roadmap and Phasing

Nigeria's implementation approach must be phased to ensure that AI-in-education is introduced in a responsible, inclusive, and scalable way. The roadmap progresses from foundational readiness to large-scale transformation by sequencing reforms in infrastructure, teacher capacity, content development, governance, and financing.

Phase 1: Foundational Readiness (Year 1–2)

The initial phase focuses on preparing the enabling environment required for AI integration. Priority actions include establishing national governance arrangements for AI-in-education, defining ethical safeguards, and upgrading baseline infrastructure in priority schools. Teacher orientation begins with core digital and AI literacy, supported by low-bandwidth and offline-capable tools that suit Nigeria's connectivity realities. National monitoring baselines are developed, and pilot programmes target secondary schools and a select number of TVET institutions and state-level training centres. These early interventions generate evidence, address risk concerns, and build confidence among teachers, parents, and school leaders.

Phase 2: Structured Rollout and Capacity Deepening (Year 3–5)

Once governance foundations and pilot learnings are consolidated, systemwide curriculum integration accelerates and teacher professional development becomes mandatory and competency-based using recognised AI teaching standards. Infrastructure expansion prioritises

rural and underserved communities, complemented by zero-rated access initiatives, community digital hubs, and safe access policies for girls and learners with disabilities. AI-enabled learning platforms are mainstreamed in secondary and tertiary programmes, while vocational institutions begin offering AI-applied modules linked to industry sectors such as agriculture, health, logistics, and creative industries. Monitoring expands from baseline indicators to real-time dashboards to guide rapid, data-driven improvement of tools and pedagogy.

Phase 3: Scale, Sustainability and Localisation (Year 6–10)

This phase focuses on achieving equitable national reach and deeper transformation of teaching and learning models. Locally developed AI solutions replace heavy dependence on imported tools, supported by sustainable financing, including public budgeting, regulated PPP models, and innovation challenge funds tied to education outcomes. Schools adopt a full spectrum of AI applications intelligent tutoring, predictive analytics for learning support, adaptive assessment, and personalised pathways, ensuring continuous learning for every child. Governance evolves into a long-term accountability structure covering algorithmic audits, data rights, digital safety (especially for girls), and responsible procurement. Lifelong learning is enhanced by strong AI pathways in adult education, workplace apprenticeships, and community learning centres.

Conclusion

Nigeria stands at a pivotal moment where Artificial Intelligence can help close longstanding gaps in access, teacher capacity, curriculum relevance, and learning outcomes across the education system. The findings of this report show that national policies and pilots have begun laying a foundation for systemwide AI integration, especially with initiatives such as curriculum reforms, teacher training, and generative-AI tutoring pilots that have already demonstrated measurable gains for students. However, progress remains uneven, with challenges in digital infrastructure, governance, equitable participation, and teacher readiness particularly evident in rural schools, non-formal education, and among girls and marginalized groups. These disparities underline the critical need for an AI-in-education strategy that centres on inclusion, safeguards, and human-led pedagogy to ensure that AI becomes a tool for narrowing rather than widening Nigeria's education inequalities.

The path forward requires deliberate alignment of curriculum, teacher development, infrastructure investments, ethical standards, and sustained partnerships across government, industry, and civil society. Strengthening data governance, enhancing local content development, and supporting schools with reliable connectivity and devices will be essential for scaling successful models beyond isolated pilots. As illustrated through relatable examples of Nigerian students using AI in *The Master not the Machine*, AI must be introduced as a supportive tool that enhances—not replaces—human learning and agency. If implemented with accountability and inclusive design, AI can support Nigeria in transforming classrooms into future-ready learning environments empowering teachers, personalizing learning for every child, and equipping youth and adults with the digital and AI competencies required in a rapidly evolving global economy.

References

Abdullahi, A. M., & Abdullahi, S. A. (2024). AI in Education: A Review of Personalized Learning and Educational Technology. ResearchGate.

https://www.researchgate.net/publication/390163119_AI_in_education_A_review_of_personalized_learning_and_educational_technology

Abimbola Makanjuola Agbola and Idakwoji Solomon. (2023.). An analysis of the role of artificial intelligence in Nigeria's educational system. [PDF]. <https://earnia.org/e4e356ef6f4f9e939e06f04e06f60ef3ef0560f0de6/AN%20ANALYSIS%20OF%20THE%20ROLE%20OF%20ARTIFICIAL%20INTELLIGENCE%20IN%20NIGERIA1.pdf> Earnia

Adebayo, S. M. (2024). Artificial Intelligence in Education: A Pathway to Improved Learning Outcomes in Nigeria. Educational Perspectives Journal of Nigeria. <https://educationalperspectives.org.ng/read.php?id=261&>

Adebayo, T. (2024, October 9). Expert Advocates AI-Driven Curriculum to Transform Nigeria's Education System. Premium Times Nigeria. <https://www.premiumtimesng.com/news/top-news/787060-expert-advocates-ai-driven-curriculum-to-transform-nigerias-education-system.html>

Adeniyi, A., & Bello, K. S. (2024). Exploring AI-Driven Adaptive Learning Systems in Nigerian Classrooms. World Journal of Innovative Management and Technology (WJIMT), 8(5), 188–198.* International Institute of Academic Research and Development (IIARD).

<https://www.iiardjournals.org/get/WJIMT/VOL.%208%20NO.%205%202024/Exploring%20AI-Driven%20Adaptive%20188-198.pdf>

Adeyemi Musliyu Adeyinka, Rabilu Abdu Yahaya, Sidiku Jegede, and Victoria Oreka. (2025.). The role of artificial intelligence in enhancing adult education in Nigeria: Technology & pedagogy for best practice and sustainable development in Nigeria. https://www.researchgate.net/publication/388707917_THE_ROLE_OF_ARTIFICIAL_INTELLIGENCE_IN_ENHANCING_ADULT_EDUCATION_IN_NIGERIA_Technology_Education_under_the_Theme_%27%27Innovative_Technology_and_Pedagogy_for_Best_Practice_and_Sustainable_Development_in_N

Ajala, O. (2025, August 18). How Poor Infrastructure and Skill Gap Hinder AI Education in Technology Education. BusinessDay Nigeria – Interview Series.

<https://businessday.ng/interview/article/how-poor-infrastructure-skill-gap-hinder-ai-education-technology-education-ajala/>

Amdan, M. A. B., Janius, N., Saidin, M. S. B., & Kasdiah, M. A. H. B. (2025). Impact of Artificial Intelligence in TVET and STEM Education among Higher Learning Students in Malaysia. Journal of Research in Mathematics, Science, and Technology Education, 2(1), 1–14. <https://doi.org/10.70232/jrmste.v2i1.15>. <https://spm-online.com/jrmste/index.php/journal/article/view/15>

Arise News. (2024, August 15). CSEA: Nigeria's AI Market to Hit \$434.4 Million by 2026 but Adoption Remains Uneven. Arise Television – Digital News.

<https://www.arise.tv/csea-nigerias-ai-market-to-hit-434-4m-by-2026-but-adoption-remains-uneven/>

Arise News. (2024, September). UBEC, US Firm to Roll Out 4.7 Million Laptops and Tablets across Nigeria in Landmark Digital Education Project. Arise TV News.

<https://www.arise.tv/ubec-us-firm-to-roll-out-47-million-laptops-tablets-across-nigeria-in-landmark-digital-education-project/>

Australian Department of Education. (2025). China's Approach to AI Education in Primary and Secondary Schools. Canberra: Department of Education. <https://www.education.gov.au/download/19495/chinas-approach-ai-education-primary-and-secondary-schools/41945/document/pdf>

Bali, B., Garba, E. J., Ahmadu, A. S., et al. (2024). Analysis of emerging trends in artificial intelligence for education in Nigeria. Discover Artificial Intelligence, 4(1), Article 110. Springer. <https://doi.org/10.1007/s44163-024-00163-y> SpringerLink+1

Banya Global. (2023, May). Nigeria GDD Brief: Digital technologies can drive development. https://banyanglobal.com/wp-content/uploads/2023/08/Nigeria-GDD-Brief_Final-508-May-2023.pdf SDI Article+1

Banya Global. (2023, May). Nigeria Gender Digital Divide (GDD) Brief – Final Report. Banya Global Development Consulting. https://banyanglobal.com/wp-content/uploads/2023/08/Nigeria-GDD-Brief_Final-508-May-2023.pdf

Bello, H., & Salihu, A. S. (2025). Artificial Intelligence

and Pedagogical Innovation in Sub-Saharan Africa. International Research Journal of Modern Science. https://www.irjms.com/wp-content/uploads/2025/04/Manuscript_IRJMS_02243_WS.pdf

BusinessDay Nigeria. (2024?). Over 50% of Nigeria's population lack digital skills — World Bank. <https://businessday.ng/technology/article/over-50-of-nigerias-population-lack-digital-skills-world-bank/>

BusinessDay Nigeria. (2025, April 22). New Secondary School Curriculum to Include AI, Coding and Journalism from September 2025. BusinessDay Media Ltd.

<https://businessday.ng/education/article/new-secondary-school-curriculum-to-include-ai-coding-journalism-from-september-2025/>

California State University – GenAI Community. (2024). Ethical Principles and AI Framework for Higher Education. California State University System. <https://genai.calstate.edu/communities/faculty/ethical-and-responsible-use-ai/ethical-principles-ai-framework-higher-education>

China State Council Information Office (SCIO). (2025, May 13). China Mandates AI Education in Primary and Secondary Schools. Beijing: SCIO. http://english.scio.gov.cn/pressroom/2025-05/13/content_117871666.html

Cisco & Carnegie Mellon University (CMU). (2024). AI and the Workforce in Africa. Cisco Systems White Paper.

https://newsroom.cisco.com/c/dam/r/newsroom/pdfs/Cisco-CMU-Whitepaper_AI-and-the-Workforce-in-Africa.pdf

Clarity Innovations / ISTE. (2022). High School Project-Based Learning Unit about AI and Society. Portland: Clarity Innovations. <https://www.clarity-innovations.com/portfolio/international-society-technology-education/high-school-project-based-learning-unit-about>

Coalition for Networked Information (CNI). (2024). Scaling Digital Literacy in Nigeria: Kolibri's Journey. Digital Curation Program, CNI.

<https://www.cni.org/topics/digital-curation/scaling-digital-literacy-in-nigeria-kolibris-journey>

Convergence AI. (2024, January 9). Leveraging PPPs for African AI growth. ConvergenceAI. <https://convergenceai.io/leveraging-ppps-for-african-ai-growth/> convergenceai.io

Critical Links / UNESCO. (2025, April). K–12 AI Curricula: UNESCO Global Guidelines. Lisbon: Critical Links. <https://critical-links.com/wp-content/uploads/2025/>

04/K-12-AI-Curricula-UNESCO.pdf

Cynthia Tysick (2025.). Scaling digital literacy in Nigeria: Kolibris' journey. <https://www.cni.org/topics/digital-curation/scaling-digital-literacy-in-nigeria-kolibris-journey>

Dalberg Advisors. (2024). Transforming Classrooms: The Growing Impact of Artificial Intelligence in African Education. Nairobi: Dalberg Global Development Advisors. <https://dalberg.com/our-ideas/transforming-classrooms-the-growing-impact-of-artificial-intelligence-in-african-education/>

Day of AI Consortium (MIT RAISE). (2024). Project-Based Learning with AI. AlxEducation Substack. <https://aixeducation.substack.com/p/project-based-learning-with-ai>

Ecofin Agency. (2024, August 13). Africa's AI Market Poised to Reach \$16.5 Billion by 2030, Mastercard Reports. Ecofin Agency Digital News.

<https://www.ecofinagency.com/news-digital/1308-48038-africa-s-ai-market-poised-to-reach-16-5-billion-by-2030-mastercard-reports>

EdTech Hub. (2025, July 23). How Education Implementers Are Approaching AI Content Generation, Curation, and Integration. EdTech Hub Policy Insights Series.

<https://edtechhub.org/2025/07/23/how-are-education-implementers-approaching-ai-content-generation-curation-and-integration/>

Education Today. (2024, July 15). UNESCO flags gender gap in STEM: Women only 35% of graduates, just 26% in AI workforce. Education Today News (UNESCO Coverage).

<https://www.educationtoday.co/news/daily-news/unesco-flags-gender-gap-in-stem-women-only-35-of-graduates-just-26-in-ai-workforce>

Efuetanu, V. (2025, June 15). LinkedIn Post – New Partnership to Advance Digital Skills for Girls in Nigeria. LinkedIn Posts.

https://www.linkedin.com/posts/vwakpor-efuetanu-2b7274240_we-just-signed-a-partnership-that-will-shape-activity-7373266859598622720-C-xc

Enrollify. (2024, May 12). Ethical Considerations for AI Use in Education. Enrollify Blog. <https://www.enrollify.org/blog/ethical-considerations-for-ai-use-in-education>

Exwhyzee News. (2024, August 28). Federal Government Introduces Artificial Intelligence Training for 6,000 Educators. Exwhyzee Nigeria.

https://exwhyzee.ng/Bpg_info/28/-Federal-Government-introduces-artificial-intelligence-training-for-6-000-educators

Federal Ministry of Communications, Innovation and Digital Economy (FMCIDE). (2025, April 24). Nigeria launches landmark AI model powered by AWARRI to advance languages and AI at scale.

<https://fmcide.gov.ng/nigeria-launches-landmark-ai-model-powered-by-awarri-to-advance-languages-and-ai-at-scale/>

Federal Ministry of Education (FME). (2013). National Policy on Education (6th Edition). Federal Government of Nigeria.

https://rtei.okfn.org/documents/National_Policy_on_Education_2013.html

Federal Ministry of Education (Nigeria). (2019, December). Policy Guidelines for Mass Literacy, Adult & Non-Formal Education. Federal Republic of Nigeria.

<https://education.gov.ng/wp-content/uploads/2019/12/POLICY-GUIDELINES-FOR-MASS-LITERACY-ADULT-NON-FORMAL-EDUCATION.pdf>

Federal Ministry of Information (FMINO). (2025, May 3). AI, Coding and Robotics: Minister of Education Calls for Structured Integration in Nigeria's Curriculum. Federal Ministry of Information, Nigeria.

<https://fmino.gov.ng/ai-coding-and-robotics-minister-of-education-calls-for-structured-integration-in-nigerias-curriculum/>

Federal Ministry of Innovation, Science & Technology (FMIST). (2024). National Artificial Intelligence Strategy (Draft). National Center for Artificial Intelligence & Robotics (NCAIR), NITDA.

https://ncair.nitda.gov.ng/wp-content/uploads/2024/08/National-AI-Strategy_01082024-copy.pdf

Frontiers in Human Dynamics. (2024). Ethical AI Decision-Making in Education: The Human Perspective. Frontiers Media S.A., 4, Article 1421273.

<https://www.frontiersin.org/journals/human-dynamics/articles/10.3389/fhumd.2024.1421273/full>

Future of Being Human Institute. (2024). AI and the Future of Human Learning. Bristol: University of Bristol Press.

<https://futureofbeinghuman.com/api/v1/file/2737ea4a-2ca4-4cb5-85cb-4c9ad88f6204.pdf>

Girls Who Code & Equal Measure. (2024). Girls Who Code Impact Evaluation Study – Final Report. Girls Who Code Inc.

https://girlswhocode.com/assets/downloads/craft-rod/downloads/GWC_Impact_Evaluation_Study_Final_Report.pdf

Girls Who Code. (2024). 2024 Annual Report. Girls Who Code Inc.

<https://girlswhocode.com/2024report/>

Global Admissions. (2024.). How universities used AI to revolutionize higher education experiences. <https://www.globaladmissions.com/blog/how-universities-used-ai-to-revolutionize-higher-education-experiences>

GlobalGiving. (2024). Digital Skills for 200 Nigerian Girls Project. GlobalGiving Foundation.

<https://www.globalgiving.org/projects/digital-skills-for-200-nigerian-girls/>

Google for Education. (n.d.). Ivy Tech case study. https://edu.google.com/intl/ALL_us/resources/customer-stories/ivytech-gcp/

Holmes, W., Bialik, M., & Fadel, C. (2022). Ethical Principles for Artificial Intelligence in Education. ResearchGate Working Paper.

https://www.researchgate.net/publication/364325080_Ethical_principles_for_artificial_intelligence_in_education

Hurix Digital. (2024, April 2). Four Proven Metrics to Assess AI's Impact on Curriculum Design. Hurix Learning Blog.

<https://www.hurix.com/blogs/four-proven-metrics-to-assess-ais-impact-on-curriculum-design/>

iAfrica News. (2024, August 12). Bildup AI Launches NGN500 Million "AI in Every Classroom" Initiative in Nigeria. iAfrica.com – Digital Economy Section.

<https://iafrica.com/bildup-ai-launches-ngn500m-ai-in-every-classroom-initiative-in-nigeria/>

ICTworks Editors. (2023, March 10). Artificial Intelligence Principles for the United Nations. ICTworks – Digital Development Blog.

<https://www.ictworks.org/artificial-intelligence-principles-united-nations/>

IIARD Journals. (2025). The Impact of AI on STEM Education and Gender. International Journal of Computer Science and Modern Technology (IJCSMT), 11(6), 74–86.*

<https://iiardjournals.org/get/IJCSMT/VOL.%2011%20NO.%206%202025/The%20Impact%20of%20AI%20on%20STEM%2074-86.pdf>

IIARD. (2025). Artificial Intelligence Awareness. WJIMT, 9(5), 101–114.

<https://iiardjournals.org/get/IJASMT/VOL.%2011%20NO.%206%202025/STEM%20Education%20and%20Gender%2043-55.pdf>

IIEP-UNESCO. (2023). Gender desks: The frontline of girls' education in Nigeria. UNESCO International

Institute for Educational Planning (IIEP).

<https://www.iiep.unesco.org/en/articles/gender-desk-s-frontline-girls-education-nigeria>

Innovation Support Network (ISN) Hubs. (2025, April 24). Multilingual Learning in Nigeria. ISN Hubs Blog.

<https://isnhubs.org.ng/2025/04/24/multilingual-learning-in-nigeria/>

International Institute of Academic Research and Development (IIARD). (2025). STEM Education and Gender. *IJASMT*, 11(6), 43–55.

<https://iiardjournals.org/get/IJASMT/VOL.%2011%20NO.%206%202025/STEM%20Education%20and%20Gender%2043-55.pdf>

International Labour Organization / International Telecommunication Union (2024). AI for Good – AI Skills Coalition. Geneva: ITU.

<https://aiforgood.itu.int/ai-skills-coalition/>

International Telecommunication Union (ITU). (2024, November 10). Measuring Digital Development: Facts and Figures 2024 – The Gender Digital Divide. ITU Statistics Report Series.

<https://www.itu.int/itu-d/reports/statistics/2024/11/10/ff24-the-gender-digital-divide/>

International Telecommunication Union (ITU). (2025, May). The AI Skills Coalition: Democratizing AI Education. Geneva: ITU.

<https://www.itu.int/hub/2025/05/the-ai-skills-coalition-democratizing-ai-education/>

ISN Hubs. (2025, April 24). Multilingual learning in Nigeria.

<https://isnhubs.org.ng/2025/04/24/multilingual-learning-in-nigeria/>

ISTE & EDUCAUSE. (2020). AI Guidelines for K–12 Education. Washington D.C.: International Society for Technology in Education.

https://cdn.iste.org/www-root/Libraries/Documents%20&%20Files/Artificial%20Intelligence/AIGDSE_1120.pdf

Kevin Mwangi. (2025.). Challenges African EdTech Startups Face in Mobile Learning. TechInAfrica.com.

<https://www.techinafrica.com/challenges-african-ed-tech-startups-face-in-mobile-learning/>

Khan, A., & Suleiman, O. (2025). Generative Artificial Intelligence (AI) and the Future of Education: Challenges and Opportunities in the Global South. *SSRN Electronic Journal*. Elsevier Social Science Research Network.

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5117653

Khanmigo. (2025.). Khanmigo AI Learning Platform. <https://www.khanmigo.ai/>

Kuni Tyessi (2025.). UBEC & U.S. firm to roll out 4.7 million laptops/tablets across Nigeria in landmark digital education project.

<https://www.arise.tv/ubec-us-firm-to-roll-out-4-7-million-laptops-tablets-across-nigeria-in-landmark-digital-education-project/>

Kurbalija, J. (2023, April 27). Ethics and AI (Part 4): The UN and Global Norms on Artificial Intelligence. DiploFoundation Blog.

<https://www.diplomacy.edu/blog/ethics-and-ai-part-4/>

Lau, J., & Lee, K. (2023). Exploring the Integration of AI Tools in Classroom-Based Learning. *Journal of Computer Assisted Learning*, 39(6), 1458–1478.

Wiley Online Library.

<https://onlinelibrary.wiley.com/doi/10.1111/jcal.12892>

Liang, T. H. (2024). Artificial Intelligence for K–12 Education: Challenges and Prospects. *Multimodal Technologies and Interaction*, 9(8), 84. MDPI.

<https://www.mdpi.com/2414-4088/9/8/84>

M.A. Oyebamiji, and Dr. Ezeala, J.I. (2023.). Digital literacy and its implications for sustainable adult education in the 21st century. *International Journal of Research and Innovation in Social Science*.

<https://rsisinternational.org/journals/ijriss/articles/digital-literacy-and-its-implications-for-sustainable-adult-education-in-the-21st-century/>

Ministry of Education, Singapore (MOE). (2025, October 7). Singapore teachers embrace digital technologies and benefit from strong professional development – OECD TALIS 2024 Study. Government of Singapore.

<https://www.moe.gov.sg/news/press-releases/2025-1007-singapore-teachers-embrace-digital-technologies-and-benefit-from-strong-professional-development-oecd-talis-2024-study>

MTN Foundation. (2024). MTN Foundation Completes the Renovation of 29 Laboratories across Nigeria's Northeast and Southeast. MTN Nigeria Foundation News.

<https://www.mtn.ng/newsfoundation/mtn-foundation-completes-the-renovation-of-29-laboratories-a-cross-nigerias-northeast-and-southeast/>

Muhammad, A. I. (2024). An Assessment of the Applications of Artificial Intelligence in Literacy Improvement Programs: A Case Study of Adult Education Centres in Jigawa State. Data Project NG.

<https://www.dataprojectng.com/project/12577/An%20Assessment%20of%20the%20Applications%20of%20Artificial%20Intelligence%20in%20Literacy%20Improvement%20Programs%3A%20A%20Case%20Study%20>

20of%20Adult%20Education%20Centers%20in%20Jig
awa%20State

Mohammed Tukur, Garba Umar Bebeji, and Muhammad Muhammad Suleiman. (2021.). The Impact of ICT in Adult and Non-Formal Education in Nigeria: A Review. *International Journal of Business, Law, and Education* Volume 2, Number 3, <https://pdfs.semanticscholar.org/30ae/0d0e30ab7e727b4920f6aefdc390b2e71a9.pdf>

Nadia Neveu (2025.). Nigeria – EdTech Startups Directory. <https://www.edtechstartups.org/country/Nigeria/>

National Board for Technical Education (NBTE). (2024, January). National TVET Policy and Strategy on ODFeL. Federal Republic of Nigeria. <https://www.nbte.gov.ng/nbte/sites/default/files/2024-01/National%20TVET%20Policy%20and%20Strategy%20on%20ODFeL.pdf> nbte.gov.ng

National Information Technology Development Agency (NITDA). (2023, July). Digital Literacy Framework. Federal Republic of Nigeria. <https://nitda.gov.ng/wp-content/uploads/2023/07/Digital-Literacy-Framework.pdf>

Nature Scientific Reports. (2025). AI-driven personalized education and gender differences in learning outcomes. Nature Publishing Group. <https://www.nature.com/articles/s41598-025-08018-x>

Nigeria Communications Commission (NCC). (2025.). SERVICOM Charter / Departmental Charters – Universal Service Provision Secretariat Service Charter. <https://ncc.gov.ng/about/servicom-charter/departmen-tal-charters/universal-service-provision-secretariat-service-charter>

Nigeria Computer Emergency Response Team (NgCERT). (2023). Nigeria Data Protection Act 2023. Federal Republic of Nigeria. https://cert.gov.ng/ngcert/resources/Nigeria_Data_Protection_Act_2023.pdf

Nigeria Data Protection Commission (NDPC). (2024). About the NDPC. Federal Government of Nigeria. <https://ndpc.gov.ng/>

Nigeria Education News. (2025, July 1). The Silent Breakdown in Nigeria's Classrooms: No Labs, No Libraries, No Digital Tools. Nigeria Education News Online. <https://thenigeriaeducationnews.com/2025/07/01/the-silent-breakdown-in-nigerias-classrooms-no-labs-no-libraries-no-digital-tools/>

Nigerian Communications Commission (NCC). (2024).

Universal Service Provision Secretariat Service Charter. NCC Departmental Charters.

<https://ncc.gov.ng/about/servicom-charter/departmen-tal-charters/universal-service-provision-secretariat-service-charter>

Njoku, U. M., Ezeoke, F. O., & Okoye, U. S. (2023). Public-Private Partnership (PPP) Implementation as Correlates for Improved Primary Education Funding in Abia State, Nigeria. *Nigerian Journal of Education Administration & Planning (NJEAP)*, 23(3). <https://njeapjournal.com/wp-content/uploads/2024/10/3-Public-Private-Partnership-PPP-Implementation-as-Correlates-for-Improved-Primary-Education-Funding-in-Abia-State-Nigeria.pdf> njeapjournal.com

Odeyemi, J., & Okonkwo, T. (2025). Artificial Intelligence Awareness and Utilization in Nigerian Universities. *World Journal of Innovative Management and Technology*, 9(5), 101–114. *International Institute of Academic Research and Development (IIARD)*. <https://iiardjournals.org/get/WJIMT/VOL.%209%20NO.%205%202025/Artificial%20Intelligence%20Awareness%20101-114.pdf>

OECD. (2024, May). OECD Updates AI Principles to Stay Abreast of Rapid Technological Developments. Paris: OECD. <https://digitalpolicyalert.org/ai-rules/2024-update-OECD-principles> Digital Policy Alert+1

Okafor, C. E. (2025). Bridging the Digital Divide: How the Nigerian Government Can Facilitate Technology Adoption in Education. *ResearchGate Preprint*. https://www.researchgate.net/publication/390666153_Bridging_the_Digital_Divide_How_the_Nigerian_Government_Can_Facilitate_Technology_Adoption_in_Education

Olaniwun Ajayi LP (Tech Practice Group). (2024, September). Overview of the Draft National Artificial Intelligence Strategy – September 2024 (OALP Tech Newsletter). Olaniwun Ajayi LP.

<https://www.olaniwunajayi.net/blog/wp-content/uploads/2024/09/Overview-of-the-Draft-National-Artificial-Intelligence-Strategy-September-2024-OALP-Tech-Newsletter.pdf>

Olaoye, F. M., & Okon, A. E. (2025). Artificial Intelligence in Education: Enhancing Digital Pedagogies in Nigerian Schools. *Research International Journal of Education and Social Sciences (RIJESSU)*, 4(4), 007–009. <https://rijessu.com/wp-content/uploads/2025/09/07-RIJE-2025-V4-N04-009.pdf>

Olasupo, A. (2025). The Master not the Machine:

Lessons in AI Literacy. Retrieved from <https://drive.google.com/file/d/1qGjrtkp1Z2n25A2i8XNBCpZUqw9L7xW1/view>

Olatunji, I. D. (2020). Panel 3 – Artificial Intelligence in Education: National Perspectives. In Federal Ministry of Education (FME). Nigeria Education Sector Panel Contributions.

<https://education.gov.ng/wp-content/uploads/2020/01/Ife-Olatunji-David-Contributor-Panel-3.pdf>

Oluwafemi, T. E. (2024). AI in Education Policy Implementation in Nigeria. *Journal of Infrastructure, Policy and Development*, 8(4). EN Press Publisher.

<https://systems.enpress-publisher.com/index.php/jipd/article/viewFile/9949/5250>

Onah, C. E., & Odo, I. S. (2025). Generative AI Adoption and the Politics of Educational Governance in Nigeria. *World Journal of Innovative Management and Technology*, 9(8), 99–111. International Institute of Academic Research and Development (IIARD).

<https://iiardjournals.org/get/WJIMT/VOL.%209%20NO.%208%202025/GENERATIVE%20AI%20ADOPTION%2099-111.pdf>

Open Edition Journals. (2024). *Revue Internationale d'Éducation de Sèvres – AI and Education Policy Comparative Insights*. France: CNDP/Open Edition.

<https://journals.openedition.org/ries/17323>

Organisation for Economic Co operation and Development (OECD). (2019). *OECD AI Principles*. Paris: OECD. <https://oecd.ai/en/ai-principles> OECD AI+1

Pavestones Legal. (2024). Analysis of Nigeria's National Artificial Intelligence Strategy. Pavestones Legal Insights.

<https://pavestoneslegal.com/analysis-of-nigerias-national-artificial-intelligence-strategy/>

Punch Newspaper (Nigeria). (2025, October 23). FG unveils AI training for 6,000 teachers nationwide. The Punch Nigeria.

<https://punchng.com/fg-unveils-ai-training-for-6000-teachers-nationwide/>

Reuters News. (2024, August 22). Nigeria's data agency fines Fidelity Bank for breaches. Reuters Africa Business.

<https://www.reuters.com/business/finance/nigerian-data-agency-fines-fidelity-bank-breaches-2024-08-22/>

Revue Internationale d'Éducation de Sèvres. (2024). Artificial Intelligence and Education: Global Comparative Insights. Open Edition Journals / CNDP France.

<https://journals.openedition.org/ries/17323?lang=es>
Right to Education Initiative. (2022, February). SDG 4 Legal Guide: Quality Education. London: RTEI. https://www.right-to-education.org/sites/right-to-education.org/files/resource-attachments/A4ID_SDG%20Legal%20Guide_SDG4_Quality%20education_February2022_EN.pdf

Rubin, C. A. (2025, August 26). AI in Education Best Practices – What's Working in Classrooms Around the Globe. Forbes Education.

<https://www.forbes.com/sites/cathyrubin/2025/08/26/ai-in-education-best-practices---whats-working-in-classrooms-around-the-globe/>

SchoolAI. (2024). How AI Helps School Admins Optimize Performance Monitoring. SchoolAI Blog.

<https://schoolai.com/blog/how-ai-helps-school-admins-optimize-performance-monitoring/>

SciencePublishingGroup. (2022). [Article Title: Education and Strategic Funding in Nigerian Context]. *Journal of Education and Social Sciences*, 12(3). <https://sciencepublishinggroup.com/article/10.11648/j.eas.20220703.12>

SEAHI Publications. (2025, April). *International Journal of Innovative Social Science & Educational Research (J-6-2025)*. SEAHI Publications.

<https://www.seahipublications.org/wp-content/uploads/2025/04/IJISSER-J-6-2025.pdf>

SEAHI Publications. (2025, July). *International Journal of Innovative Information Systems & Technology Research (S-15-2025)*.

<https://www.seahipublications.org/wp-content/uploads/2025/07/IJISTR-S-15-2025.pdf>

SEAHI Publications. (2025, July). *International Journal of Innovative Information Systems and Technology Research (IJISTR-S-15-2025)*. SEAHI Publications.

<https://www.seahipublications.org/wp-content/uploads/2025/07/IJISTR-S-15-2025.pdf>

SEAHI Publications. (2025, October). *International Journal of Innovative Information Systems & Technology Research (D-5-2025)*. SEAHI Publications.

<https://www.seahipublications.org/wp-content/uploads/2025/10/IJISTR-D-5-2025.pdf>

Silas, E. I., Ekong, M. O., & Akpan, I. M. (2024). Exploring AI-Driven Adaptive Learning Systems for Personalized Education in Nigerian TVET Institutions to Enhance Student Engagement and Skill Acquisition Outcomes. *World Journal of Innovation and Modern Technology*, 8(5), 188–198. International Institute of Academic Research and Development. <https://www.iiardjournals.org/get/WJIMT/VOL.%208%20NO.%205%202024/WJIMT-2024-05-188-198.pdf>

20NO.%205%202024/Exploring%20AI-Driven%20Adaptive%20188-198.pdf iardjournals.org+1

SkillsBuild. (2025.). Explore Learning – Artificial Intelligence (Adult Learners).
<https://skillsbuild.org/adult-learners/explore-learning/artificial-intelligence>

SyncSci. (2023). [Article Title: AERE – Public-Private Partnership in Education]. Applied Economics and Educational Review, 1(3).
<https://www.syncsci.com/journal/AERE/article/download/AERE.2023.01.003/844/>

TechCabal Insights. (2025, March 12). Strengthening Nigeria's AI Literacy Drive through Real Reform. Lagos: TechCabal.
<https://insights.techcabal.com/strengthening-nigeria-s-ai-literacy-drive-through-real-reform/>

The Case HQ. (2024, June 3). Ethical Considerations of AI in Education: Balancing Innovation and Privacy. The Case HQ Insights.
<https://thecasehq.com/ethical-considerations-of-ai-in-education-balancing-innovation-and-privacy/>

The Guardian Nigeria. (2025, October 11). More Nigerian students embracing AI, says Google data. The Guardian Nigeria Technology News.

<https://guardian.ng/technology/more-nigerian-students-embracing-ai-says-google-data/>

The Guardian. (2025, March 20). Cost of data in the developing world blocks female entrepreneurs, report finds. The Guardian (Global Development).

<https://www.theguardian.com/global-development/2025/mar/20/cost-data-developing-world-digital-women-in-business-report-female-entrepreneurs-internet-access>

This Day Live. (2025, October 23). FG launches scheme to empower teachers with mobile devices and free data. This Day Newspaper (Nigeria).

<https://www.thisdaylive.com/2025/10/23/fg-launches-scheme-to-empower-teachers-with-mobile-devices-free-data/>

U.S. Department of Education. (2023). Artificial Intelligence and the Future of Teaching and Learning: Insights and Recommendations. Washington D.C.: U.S. Department of Education.
<https://www.ed.gov/sites/ed/files/documents/ai-report/ai-report.pdf>

UK Cabinet Office. (2025). Algorithmic Transparency Record – Oak National Academy (AILA Lesson Assistant). Gov.uk.

<https://www.gov.uk/algorithmic-transparency-record/s/oak-naddtional-academy-aila-oaks-ai-lesson-assistant>

UN Digital Library. (2022). Recommendation on the Ethics of Artificial Intelligence – UNESCO Official Record. United Nations Digital Library.

<https://digitallibrary.un.org/record/4062376?ln=en>

UNESCO. (2019). Beijing Consensus on Artificial Intelligence and Education. Paris: UNESCO.
<https://unesdoc.unesco.org/ark%3A/48223/pf0000368303MOEChina+2UNESCO+2>

UNESCO. (2021). Recommendation on the Ethics of Artificial Intelligence. United Nations Educational, Scientific and Cultural Organization (UNESCO).
<https://unesdoc.unesco.org/ark%3A/48223/pf0000386693>

UNESCO. (2021). Recommendation on the Ethics of Artificial Intelligence. United Nations Educational, Scientific and Cultural Organization (UNESCO).
<https://www.unesco.org/en/artificial-intelligence/recommendation-ethics>

UNESCO. (2023). Guidance for Generative AI in Education and Research. United Nations Educational, Scientific and Cultural Organization (UNESCO).
<https://www.unesco.org/en/articles/guidance-generative-ai-education-and-research>

UNESCO. (2023). What you need to know about UNESCO's new AI competency frameworks: Students and teachers. Paris: UNESCO.
<https://www.unesco.org/en/articles/what-you-need-know-about-unescos-new-ai-competency-frameworks-students-and-teachers>

UNESCO. (2023.). How digital learning is improving livelihoods in Nigeria.
<https://www.unesco.org/en/articles/how-digital-learning-improving-livelihoods-nigeria>

UNESCO. (2024). AI Literacy and the New Digital Divide: A Global Call to Action. Paris: UNESCO Ethics of AI.
<https://www.unesco.org/ethics-ai/en/articles/ai-literacy-and-new-digital-divide-global-call-action>

UNESCO. (2024). How Digital Learning Is Improving Livelihoods in Nigeria. United Nations Educational, Scientific and Cultural Organization (UNESCO).
<https://www.unesco.org/en/articles/how-digital-learning-improving-livelihoods-nigeria>

UNESCO. (2024). UNESCO's Global Skills Academy spotlight session – WorldSkills Conference 2024.
<https://www.unesco.org/en/articles/unescos-global-skills-academy-spotlight-session-worldskills-conference-2024>

UNESCO. (2025). AI Competency Framework for Teachers. UNESCO Education Sector Report.

<https://www.unesco.org/en/articles/ai-competency-framework-teachers>

United Nations Office of the High Commissioner for Human Rights (OHCHR). (2022). UNESCO – Ethics of Artificial Intelligence: Human Rights and Governance Dimensions. OHCHR/UNESCO Report.

<https://www.ohchr.org/sites/default/files/2022-03/UNESCO.pdf>

United Nations Population Fund (UNFPA). (n.d.). Technology-Facilitated Gender Based Violence (TFGBV). <https://www.unfpa.org/TFGBV>

United Nations System Chief Executives Board for Coordination (UNSCEB). (2021). Principles for the Ethical Use of Artificial Intelligence in the United Nations System. UNSCEB Secretariat, Geneva.

<https://unsceb.org/principles-ethical-use-artificial-intelligence-united-nations-system>

United Nations System Chief Executives Board for Coordination (UNSCEB). (2021). Principles for the Ethical Use of Artificial Intelligence in the United Nations System. New York: United Nations. <https://unsceb.org/principles-ethical-use-artificial-intelligence-united-nations-system>

United States Department of Education. (2023). Artificial Intelligence in Education Report. Washington D.C.: U.S. Department of Education. <https://www.ed.gov/sites/ed/files/documents/ai-report/ai-report.pdf>

Universal Service Provision Fund (USPF). (2018). USPF Annual Report 2018 (Final Draft). https://uspf.gov.ng/images/USPF_AR_2018_Final_Draft.pdf

Universal Service Provision Fund (USPF). (2018). USPF Annual Report 2018. Federal Ministry of Communication and Digital Economy (Nigeria). https://uspf.gov.ng/images/USPF_AR_2018_Final_Draft.pdf

University of Melbourne. (2024?). University students are using AI — but not how you think. Pursuit. <https://pursuit.unimelb.edu.au/articles/university-students-are-using-ai-but-not-how-you-think> Unknown Author(s). (2022, August). Best Practices of Promoting Adult and Non-Formal Education via Distance and E-Learning in Nigeria. Ibadan Adult Education Journals. <https://ibadanadulteducationjournals.com.ng/media/2022/08/Best-Practices-of-Promoting-Adult-and-Non-Formal-Education-via-Distance-and-E-Learning-in-Nigeria.pdf>

Unknown Author(s). (2024). Title missing: SLE Journal

article on AI in education]. Smart Learning Environments. SpringerOpen. <https://slejournal.springeropen.com/articles/10.1186/s40561-024-00352-3>

U.S. Department of Education. (2023). Artificial Intelligence and the Future of Teaching and Learning: Insights and Recommendations. Washington, DC: Office of Educational Technology. Available at: <https://www.ed.gov/sites/ed/files/documents/ai-report/ai-report.pdf>

Vanguard Newspaper (Nigeria). (2025, October 21). Only 23% of rural communities have internet access in Nigeria – NCC. Vanguard Nigeria Technology. <https://www.vanguardngr.com/2025/10/only-23-of-rural-communities-have-internet-access-in-nigeria-ncc/>

World Bank & Partners. (2025?). Nigeria Digital Skills Report (final). [PDF]. <https://thedocs.worldbank.org/en/doc/a607bb6e3b76d2be0f3db8db34dcf73e-0140022025/original/1Nigeria-TFOC2441-Digital-Skills-Report-final.pdf>

World Bank Blog. (2024, June 19). Addressing the Learning Crisis with Generative AI: Lessons from Nigeria. World Bank Development Talk. <https://blogs.worldbank.org/en/developmenttalk/addressing-the-learning-crisis-with-generative-ai--lessons-from->

World Bank Blogs – Development Talk. (2024, June 19). Addressing the Learning Crisis with Generative AI: Lessons from Nigeria. World Bank Group. <https://blogs.worldbank.org/en/developmenttalk/addressing-the-learning-crisis-with-generative-ai--lessons-from->

World Bank Blogs – Education. (2024, July 3). From Chalkboards to Chatbots: Transforming Learning in Nigeria. World Bank Group. <https://blogs.worldbank.org/en/education/From-chalkboards-to-chatbots-Transforming-learning-in-Nigeria>

World Bank Blogs – Nasikiliza. (2024, March 7). Transforming Futures: Digital Skills for Girls and Women in Northern Nigeria. World Bank Group. <https://blogs.worldbank.org/en/nasikiliza/transforming-futures-digital-skills-for-girls-and-women-in-northern-nigeria>

World Bank Group. (2024). Addressing the Learning Crisis with Generative AI – Lessons from Nigeria. World Bank Education Global Practice. <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099548105192529324>

World Bank Group. (2024). Addressing the Learning Crisis with Generative AI – Lessons from Nigeria. <https://documents1.worldbank.org/curated/en/099548105192529324/pdf/IDU-c09f40d8-9ff8-42dc-b315-591157499be7.pdf>

World Bank Group. (2024). Addressing the Learning Crisis with Generative AI: Lessons from Nigeria. World Bank Education Global Practice.

<https://documents1.worldbank.org/curated/en/099548105192529324/pdf/IDU-c09f40d8-9ff8-42dc-b315-591157499be7.pdf>

World Bank Group. (2025). Innovative Financing of Education Technology (EDU Working Paper No. 19). World Bank Education Global Practice.

<https://thedocs.worldbank.org/en/doc/a607bb6e3b76d2be0f3db8db34dcf73e-0140022025/related/9EDU-WP-19-Innovative-financing-of-education-technology.pdf>

World Bank. (2024). Addressing the Learning Crisis with Generative AI: Lessons from Nigeria. Washington D.C.: World Bank Group. <https://documents1.worldbank.org/curated/en/099548105192529324/pdf/IDU-c09f40d8-9ff8-42dc-b315-591157499be7.pdf>

World Bank. (2025). Innovative Financing of Education Technology. (Working Paper 9EDU-WP-19). The World Bank. <https://thedocs.worldbank.org/en/doc/a607bb6e3b76d2be0f3db8db34dcf73e-0140022025/related/9EDU-WP-19-Innovative-financing-of-education-technology.pdf>

nology.pdf

World Economic Forum. (2024, April). Prepare for the future: Policy ideas for AI in education. Geneva: WEF. <https://www.weforum.org/stories/2024/04/prepare-future-policy-ideas-ai-in-education/>

Xu, L., & Tang, Q. (2025). Evaluating Generative AI for Secondary Education. arXiv preprint arXiv:2505.21398. <https://arxiv.org/abs/2505.21398>

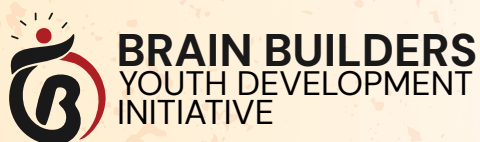
Zhang, L., & Xu, Y. (2024). Responsible AI in Education – A Comprehensive Ethical Framework. arXiv preprint arXiv:2406.18900.

<https://arxiv.org/abs/2406.18900>

Zhang, Y., & Chen, L. (2023). Developing AI Literacy: Teacher Perspectives and Pedagogical Practices. *Smart Learning Environments*, 10(15). Springer. <https://link.springer.com/article/10.1007/s40692-023-00304-9>

Zhou, X., & Li, P. (2024). Artificial Intelligence in Education: A Systematic Review of Global Trends and Emerging Issues. *Computers and Education: Artificial Intelligence*, 5, 100212. Elsevier. <https://www.sciencedirect.com/science/article/pii/S2666920X2400122X>

Zouhaier Slimi (2023). The Impact of Artificial Intelligence on Higher Education: An Empirical Study, *European Journal of Educational Sciences*, March 2023 edition Vol.10 No.1 ISSN: 1857- 6036. <https://files.eric.ed.gov/fulltext/EJ1384682.pdf>



Email: brainbuilderedu@gmail.com **Tel:** 07068775529

Nigeria Address: No. 12 T.O.S. Benson Crescent, Utako Abuja, 900108, FCT

Cameroun Address: Yaoundé- Montee Jouvence, Patmos Church Street 8.560

US Address: 1301 S Bowen Rd STE 450 Arlington TX 76013

www.thebrainbuilders.org

WITH SUPPORT FROM:



Patrick J McGovern
FOUNDATION